

Revised MTech (Artificial Intelligence) Teaching Scheme

To be applicable with effect from 2025



Centre for Artificial Intelligence
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Vision

To become a center of excellence in artificial intelligence education, research, and innovation leading to knowledge advancement that translates into significant social and economic welfare

Mission

- M1: To become a national access point for artificial intelligence education, competence, and infrastructure.**
- M2: To facilitate national, and international cooperation and collaboration in artificial intelligence.**
- M3: To encourage the use of artificial intelligence in an ethical, transparent, and inclusive manner that fosters creativity, innovation, and an entrepreneurial mindset**

Program Educational Objectives (PEOs)

- PEO1: Develop expertise in analysing, designing, and formulating complex research problems by applying in-depth domain knowledge and research principles in AI and allied fields.
- PEO2: Prepare for a successful career in academia, research, entrepreneurship, or AI-driven industries by mastering cutting-edge AI technologies, tools, and methodologies.
- PEO3: Work effectively in diverse, multidisciplinary, and multicultural environments while adhering to professional ethics, societal norms, and global AI regulations.
- PEO4: Engage in continuous learning to stay updated with evolving AI advancements while contributing to technological progress for societal and national development.

Program Outcomes (POs)

- PO1: Demonstrate the ability to independently conduct research, investigate challenges, and develop innovative AI-based solutions for real-world applications.
- PO2: Exhibit proficiency in writing and presenting substantial technical reports, research papers, and project documentation in the field of AI.
- PO3: Attain an advanced level of expertise in deep learning applications in various domains, surpassing the knowledge level required at the undergraduate level.
- PO4: Apply AI principles with a strong emphasis on ethics, fairness, and responsible AI deployment, ensuring societal and industrial impact while adhering to professional standards.

Program Specific Outcomes (PSOs)

- PSO1: Ability to apply mathematical, statistical concepts and programming skills to analyze and develop innovative AI solutions for the real world problems.
- PSO2: Demonstrate inferential analysis and enrich skills required to qualify for Employment, Higher studies and Research in Artificial Intelligence with ethical values.

Revised Teaching Scheme

First Semester

Sr. No.	Course Code	Course Title	L	T	P	Credits
1.	AI-501	Mathematical Foundations for AI	2	0	0	2
2.	AI-503	Computer Programming with Python	2	0	0	Non-Credit
3.	AI-505	Advanced Data Structures and Algorithms	3	0	0	3
4.	AI-507	Principles of Artificial Intelligence and Machine Learning	3	0	0	3
5.	AI-509	Machine Learning for Time Series Forecasting	3	0	0	3
6.	AI-6xx	Elective -I*	3	0	0	3
7.	AI-6xx	Elective -II*	3	0	0	3
8.	AI-511	AI Survey	1	0	0	Non-Credit
9.	AI-513	Industry/Expert AI Lecture Series I	1	0	0	Non-Credit
10.	AI-523	Programming with Python Laboratory	0	0	3	2
11.	AI-525	Advanced Data Structures and Algorithms Laboratory	0	0	3	2
12.	AI-527	Machine Learning Laboratory	0	0	3	2

Total Credits: 23

Second Semester

Sr. No.	Course Code	Course Title	L	T	P	Credits
1.	AI-502	Computer Vision and Natural Language Processing	3	0	0	3
2.	AI-504	Foundations of Deep Learning	3	0	0	3
3.	AI-506	Concurrent Programming	3	0	0	3
4.	AI-508	Reinforcement Learning	3	0	0	3
5.	AI-6xx	Elective -III*	3	0	0	3
6.	AI-6xx	Elective -IV*	3	0	0	3
7.	AI-510	Industry/Expert Lecture Series II	1	0	0	Non-Credit
8.	AI-512	AI Design	1	0	0	Non-Credit
9.	AI-514	Artificial Intelligence Governance	1	0	0	Non-Credit
10.	AI-524	Deep Learning Laboratory	0	0	3	2
11.	AI-526	Concurrent Programming Laboratory	0	0	3	2

Total Credits: 20

Third Semester

Sr. No.	Course Code	Course Title	L	T	P	Credits
1.	AI-515	Independent Study Seminar	3	0	0	3
2.	AI-600	Dissertation Phase 1	16	0	0	16

Total Credits: 19

Fourth Semester

Sr. No.	Subject Code	Subject Name	L	T	P	Credits
1.	AI-600	Dissertation Phase 2	18	0	0	18

Total Credits: 18

Total Program Credits = 80

Every student will opt for one specialization stream and all the four Electives I-IV have to be from that opted stream only

Specialization Stream I: Cloud Computing for AI

Sr. No.	Course Code	Course Title	L	T	P	Credits	Elective
1.	AI-601	Cloud Infrastructure for AI	3	0	0	3	I
2.	AI-603	MLOps on Cloud Platforms	3	0	0	3	II
3.	AI-602	Cloud-Native AI Development	3	0	0	3	III
4.	AI-604	Advanced Topics in Cloud AI and Responsible AI	3	0	0	3	IV

Specialization Stream II: AI for Data Analytics

Sr. No.	Course Code	Course Title	L	T	P	Credits	Elective
1.	AI-611	Advanced Data Management, Engineering, and Preprocessing for AI	3	0	0	3	I
2.	AI-613	Advanced Machine Learning Models for Data Analytics	3	0	0	3	II
3.	AI-612	Advanced AI for Unstructured Data and Strategic Analytics	3	0	0	3	III
4.	AI-614	Advanced Data Engineering for AI System	3	0	0	3	IV

Specialization Stream III: Generative AI

Sr. No.	Course Code	Course Title	L	T	P	Credits	Elective
1.	AI-621	Generative Artificial Intelligence Fundamentals	3	0	0	3	I
2.	AI-623	Applied Generative AI and Prompt Engineering	3	0	0	3	II
3.	AI-622	Large Language Models	3	0	0	3	III
4.	AI-624	Agentic AI	3	0	0	3	IV

Specialization Stream IV: AI for Signal and Image Processing

Sr. No.	Course Code	Course Title	L	T	P	Credits	Elective
1.	AI-631	Fundamentals of Signal and Image Processing for AI	3	0	0	3	I
2.	AI-633	Machine Learning for Signal and Image Processing	3	0	0	3	II
3.	AI-632	Deep Learning for Signal and Image Processing	3	0	0	3	III
4.	AI-634	Advanced AI applications in Signals and Image Domain	3	0	0	3	IV

First Semester

Course Code	AI-501					
Subject Name	Mathematical Foundations for AI					
Contact hours/Credit Scheme (L-T-P-C)	2-0-0-2					
Pre-requisites	None					
Course Objectives						
1. Develop a deep understanding of core mathematical tools used in AI, including linear algebra, calculus, probability, and optimization. 2. Analyse and apply mathematical reasoning to formalize AI algorithms and evaluate their behaviour. 3. Build the ability to bridge theory and implementation for AI model design and learning processes.						
Course Outcomes						
CO1.Apply linear algebra techniques (matrix operations, vector spaces, eigenvalues) in modelling AI systems.						
CO2.Utilize differential calculus and multivariate gradients in training algorithms such as backpropagation.						
CO3.Analyse probabilistic models, conditional independence, and Bayes' theorem for uncertainty handling in AI.						
CO4.Formulate optimization problems and apply methods like gradient descent and Lagrange multipliers..						
Mapping						
	PO1	PO2	PO3	PO4	PSO1	PSO2
CO1	✓			✓		✓
CO2	✓	✓	✓	✓	✓	✓
CO3	✓	✓	✓		✓	✓
CO4	✓	✓	✓	✓	✓	✓
Course Contents						
Unit 1 (5 Hours)	Introduction to Mathematics for AI. Role of mathematics in AI and machine learning. Review of sets, functions, and notation. Overview of AI models and where math fits in.					
Unit 2 (4 Hours)	Vectors, Matrices, and Linear Transformations. Vectors and vector spaces. Matrix operations and types. Linear transformations and geometric interpretations. Eigenvalues, Eigenvectors, and Spectral Theorem. Characteristic equations. Diagonalization and eigen decomposition. Applications in PCA and deep learning					
Unit 3 (5 Hours)	Norms, Distance Metrics, and Projections. Vector and matrix norms. Euclidean and Mahalanobis distances. Projections onto subspaces and least squares approximation. Calculus for AI: Derivatives and Gradients. Single-					

	variable and multivariable derivatives. Chain rule and gradient vectors. Applications in loss minimization and gradient descent.
Unit 4 (3 Hours)	Optimization Fundamentals. Convex sets and convex functions. Local vs. global minima. Gradient descent, stochastic gradient descent (SGD), and convergence. Constrained Optimization. Lagrange multipliers. KKT conditions. Applications in support vector machines and constrained learning, Markov Model.
Unit 5 (5 Hours)	Probability Theory Basics. Probability spaces, events, and conditional probability. Bayes' theorem. Random variables, PDFs, PMFs, and expectations. Distributions and Sampling. Common distributions: Bernoulli, Binomial, Gaussian, Exponential. Joint, marginal, and conditional distributions. Law of large numbers and central limit theorem
Unit 6: (3 Hours)	Information Theory for AI. Entropy, cross-entropy, and KL divergence. Mutual information. Applications in classification, compression, and reinforcement learning. Mathematical Foundations of Learning. Bias-variance tradeoff. Overfitting, underfitting, and regularization. VC dimension and generalization error (basic overview).
References	
1. Goodfellow I, Bengio Y, Courville A. <i>Deep Learning</i>. MIT Press. 2016. 2. Shalev-Shwartz S, Ben-David S. <i>Understanding Machine Learning: From Theory to Algorithms</i>. Cambridge: Cambridge University Press. 2014. 3. Axler S. <i>Linear Algebra Done Right</i>. Springer. 2015 4. Bishop CM. <i>Pattern Recognition and Machine Learning</i>. Springer. 2006. 5. Papoulis A, Pillai SU. <i>Probability, Random Variables and Stochastic Processes</i>. McGraw-Hill. 2002. 6. Cover TM, Thomas JA. <i>Elements of Information Theory</i>. Wiley-Interscience. 2006. 7. Boyd S, Vandenberghe L. <i>Convex Optimization</i>. Cambridge: Cambridge University Press. 2004. 8. Research papers from IEEE and other reputed publishing houses	

Course Code	AI-503					
Subject Name	Computer Programming with Python					
Contact hours/Credit Scheme (L-T-P-C)	2-0-0-0					
Pre-requisites	None					
Course Objectives						
1. Analyse real life situational problems and think creatively about solutions to them. 2. Apply a solution clearly and accurately in a program using Python. 3. Apply their logic to solve and code various data visualizations problems and machine learning approach to programs						
Course Outcomes						
CO1. Develop a Python program to demonstrate basics of variables and operators. CO2. Perform operations on data structures using conditions and loops in Python CO3. Develop functions for a given problem.						
Mapping						
	PO1	PO2	PO3	PO4	PSO1	PSO2
CO1	✓		✓	✓		✓
CO2	✓	✓		✓	✓	✓
CO3	✓	✓	✓		✓	✓
Course Contents						
Unit 1 (5 Hours)	Introduction to Python and Computer Programming: User vs Programmer, what is Code? Introduction to Python, Program for human, Program for Python.					
Unit 2 (4 Hours)	Data Types, Variables, Basic Input-Output Operations, Basic Operators: Your first program, Python literals, Operators - data manipulation tools, Variables - data-shaped boxes, How to talk to a computer?					
Unit 3 (5 Hours)	Boolean Values, Conditional Execution, Loops, Lists and List Processing, Logical and Bitwise Operations: Making decisions in Python, Python's loops, Logic and bit operations in Python, Lists - collections of data, Sorting simple lists - the bubble sort algorithm, Lists - some more details, Lists in advanced applications.					
Unit 4 (3 Hours)	Functions, Tuples, Dictionaries, and Data Processing: Writing functions in Python, How functions communicate with their environment? Returning a result from a function, Scopes in Python, Let's make some fun... sorry, functions, Tuples and dictionaries.					
Unit 5 (5 Hours)	Modules, Packages, String and List Methods, and Exceptions: Using modules, Some useful modules, What is package?, Errors - the programmer's daily bread, The anatomy of exception, Some of the most useful exceptions, Characters and strings vs. computers, Python's nature of strings, String methods, Strings in action, Four simple programs.					
Unit 6: (3 Hours)	The Object-Oriented Approach: Classes, Methods, Objects, and the Standard Objective Features; Exception Handling, and Working with Files: Basic					

	concepts of object programming, A short journey from procedural to object approach, Properties, Methods, Inheritance - one of object programming foundations, Exceptions once again, Generators and closures, Processing files, Working with real files.
References	
1. Thareja R. <i>Python Programming: Using Problem Solving Approach</i>, Oxford University Press, 2005 2. Brown MC. <i>Python The Complete Reference</i>. Tata Mcgraw Hill, 2007. 3. Kamthane A, Kamthane AA. <i>Programming And Problem Solving With Python</i>. Tata Mcgraw Hill, 2012 4. Severance C. <i>Python For Everybody - Exploring Data In Python 3</i>, Spd 5. Research papers from IEEE, ELSIVER, SPINGER, other journals of repute.	

Course Code	AI-505					
Subject Name	Advanced Data Structure and Algorithms					
Contact hours/Credit Scheme (L-T-P-C)	3-0-0-3					
Pre-requisites	None					
Course Objectives						
1. Help understand linear and non-linear data structures						
2. Introduce sorting, searching and hashing algorithms.						
3. Introduce dynamic data structures along with advanced data structure algorithms						
Course Outcomes						
CO1.Understand the concepts of data structures, algorithm and analyse their time complexity.						
CO2.Apply sequential data structures e.g. Array and Linked list to solve basic problems.						
CO3.Apply and analyze stack and queue data structures to solve practical problems in real-life scenarios.						
CO4.Apply variety of data structures, including trees, graphs, and hashing techniques, to address diverse and complex real-time computing problems.						
Mapping						
	PO1	PO2	PO3	PO4	PSO1	PSO2
CO1	✓	✓	✓	✓	✓	
CO2	✓		✓		✓	✓
CO3	✓	✓	✓	✓	✓	✓
CO4	✓	✓	✓	✓	✓	✓
Course Contents						
Unit 1 (7 Hours)	Algorithm Specification, Recursion, Performance Analysis, Asymptotic Notation, The Big-O, Omega and Theta Notation, Programming Style, Refinement of Coding, Time-Space Trade Off, Testing, Data Abstraction					
Unit 2 (8 Hours)	Array Address calculation, implementing basic algorithms on 1D and 2D, Singly Linked Lists, Two-way Header List, doubly linked list, Linked List in Array, Polynomial representation, and addition, Generalized linked list, Circular linked list, Garbage Collection and Compaction, Stack implementing using arrays and linked lists, implement basic operations on stack expression evaluation (prefix and postfix), Conversion of Infix to prefix and Postfix Expressions, Queue implementation using arrays and linked lists, Priority Queue, Circular Queue					
Unit 3 (9 Hours)	Trees, Tree Traversals, Binary Tree, Threaded Binary Tree, Binary Search Tree, B & B+ Tree, AVL Tree, Splay Tree. Graph, Directed, Undirected, Basic Terminologies and Representations, Graph Search and Traversal Algorithms. Heaps as priority queues, heap implementation, insertion and deletion operations, binary heaps, binomial and Fibonacci heaps, heapsort.					
Unit 4 (8 Hours)	Sequential Search, Binary Search, Breadth First Search, Depth First Search Insertion Sort, Selection Sort, Shell Sort, Divide and Conquer Sort, Merge Sort, Quick Sort, Heapsort, Radix Sort, Counting Sort.					
Unit 5 (8 Hours)	Files; Sequential, Direct, Indexed Sequential, Hashed and Various Types of Accessing Schemes. Dictionary: Abstract data type, array and tree-based implementations.					

	Hashing: Search efficiency in lists and skip lists, hashing as a search structure, hash table, collision resolution, universal hashing, linear open addressing, chains, hash tables in data-compression. LZW algorithm. Multimedia Structures: Segment trees, k-d trees, Point Quadrees, MX-Quad trees, R-trees. String Matching Algorithms: Introduction, The Brute-Force- Algorithm, Rabin-Karp Algorithm, String Matching with Finite automata, Knuth-Marries-Pratt Algorithm
References	
1. Horowitz E, Sahni S, Freed SA. <i>Fundamentals of Data Structures</i> Universities Press, Second Edition, 2008 2. Aho AV, Hopperoft JE, Ullman JD, <i>Data Structures and Algorithms</i>. Pearson Education, 1982. 3. Goodrich MT, Tamassia R, Goldwasser MH, <i>Data Structures and Algorithms in Python</i>. Wiley, 2021. 4. Lee KD, Hubbard S, <i>Data Structures and Algorithms with Python</i>. Springer, 2015 5. Brass P. <i>Advanced Data Structures</i>. Cambridge, 2008. 6. Research papers from IEEE, ELSIVER, SPINGER, other journals of repute.	

Course Code	AI-507					
Subject Name	Principles Of Artificial Intelligence And Machine Learning					
Contact hours/Credit Scheme (L-T-P-C)	3-0-0-3					
Pre-requisites	None					
Course Objectives						
1. Introduce students to the basic concepts of artificial intelligence and machine learning and illustrate basic principles of AI toward problem solving using Search Strategy. 2. Acquire knowledge on intelligent systems and agents, formalization of knowledge, reasoning, machine learning and applications at basic level. 3. Enable the students to apply and implement these techniques in applications which involve perception, reasoning and learning with respect to AI and ML.						
Course Outcomes						
CO1. Understand the concept of Artificial Intelligence and basics of various search techniques and learning algorithms.						
CO2. Familiarize with knowledge-based AI systems and approaches.						
CO3. Apply the aspect of probabilistic approach to AI.						
CO4. Analyze unsupervised, supervised and reinforcement learning						
Mapping						
	PO1	PO2	PO3	PO4	PSO1	PSO2
CO1	✓		✓	✓		✓
CO2	✓	✓		✓	✓	✓
CO3	✓	✓	✓		✓	✓
CO4		✓	✓		✓	✓
Course Contents						
Unit 1 (7 Hours)	Introduction to Artificial Intelligence: What Does AI Mean? Reviewing Basic Definitions and AI History, Types of AI, Size of AI Market and Growth Rate, The birth of AI, Applications of AI, Intelligent Agents: Agents and Environment, Structure of Agents, AI problem-solving approaches: search algorithms, problem spaces, Uninformed Search Strategies – Depth-first search – Breadth-first search – Uniform-cost search, Informed (Heuristic) Search Strategies – Heuristic Functions – Greedy best-first search – A* search, Hill climbing search, Adversarial Search					
Unit 2 (8 Hours)	Knowledge Representation and Reasoning: Representing knowledge in AI systems: propositional logic, predicate logic - Logical reasoning and inference techniques - Semantic networks, frames, and ontologies, resolution, conjunctive normal form, semantics of propositional logic, Wumpus world, Inference in semantic networks,					

Unit 3 (6 Hours)	Representing and Reasoning with Uncertain Knowledge: Probability concepts, connection to logic, independence, Bayes rule, Bayesian networks, Gaussian Bayesian networks, probabilistic inference, random variables, discrete random variables, Expectations, medians and variance of random variables, Covariance, sample applications
Unit 4 (8 Hours)	Introduction to Machine Learning: Overview of machine learning: supervised learning, Regression and classification algorithms - Model evaluation and validation techniques, Representative learning algorithms. Learning nearest neighbour, naive Bayes and its variants, neural networks and their variants, and decision trees and their variants unsupervised learning, reinforcement learning, classification of machine learning algorithms, K-means clustering - Hierarchical clustering,
Unit 5 (6 Hours)	Modules, Packages, String and List Methods, and Exceptions: Using modules, Some useful modules, What is package?, Errors - the programmer's daily bread, The anatomy of exception, Some of the most useful exceptions, Characters and strings vs. computers, Python's nature of strings, String methods, Strings in action, Four simple programs.
Unit 6: (5 Hours)	Deep Learning Fundamentals and Natural Language Processing: Introduction to artificial neural networks (ANNs) - Deep feedforward networks and activation functions - Training neural networks: backpropagation algorithm, Basics of text processing and tokenization - Named Entity Recognition (NER) and Part-of-Speech (POS) tagging - Sentiment analysis and text classification, - Basics of CNN architecture, Introduction to RNNs and Long Short-Term Memory (LSTM) networks, Applications of RNNs in natural language processing and time series analysis
References	
1. Winston PH. <i>Artificial Intelligence</i>, Addison-Wesley Publishing Company, 2004. 2. Nilsson NJ. <i>Principles of Artificial Intelligence</i>. Springer Heidelberg, 2014. 3. Russell S, Norvig P. <i>Artificial Intelligence: A Modern Approach</i>. PHI 2009. 4. Nilsson NJ. <i>Quest for Artificial Intelligence</i>. Cambridge University Press, 2010. 5. Mitchell TM. <i>Machine Learning</i>. Mc Graw Hill Education, 1997. 6. Alpaydin E, <i>Introduction to Machine Learning</i>. MIT Press, 2014. 7. Flaszinski M. <i>Introduction to Artificial Intelligence</i>. SPRINGER 2017. 8. Clark RM. <i>Intelligence analysis</i>. Sage Publications, 2012. 9. Research papers from IEEE, ELSIVER, SPINGER, other journals of repute.	

Course Code	AI-509					
Subject Name	Machine Learning for Time Series Forecasting					
Contact hours/Credit Scheme (L-T-P-C)	3-0-0-3					
Pre-requisites	None					
Course Objectives						
1. Introduce the fundamentals of time series data and classical forecasting models.						
2. Teach preprocessing and feature engineering specific to temporal data.						
3. Transition to machine learning and deep learning approaches for time series.						
4. Expose students to state-of-the-art AI architectures for time series forecasting and classification.						
5. Enable students to evaluate and deploy AI models for real-world time series applications.						
Course Outcomes						
CO1. Understand and manipulate univariate and multivariate time series data.						
CO2. Apply classical time series models.						
CO3. Engineer temporal features and apply relevant data transformations.						
CO4. Build AI models for forecasting and anomaly detection.						
Mapping						
	PO1	PO2	PO3	PO4	PSO1	PSO2
CO1	✓	✓	✓		✓	✓
CO2		✓	✓	✓		
CO3	✓	✓		✓	✓	
CO4	✓		✓		✓	✓
Course Contents						
Unit 1 (8 hours)	What is time series data? Definition and real-world examples (stock prices, sensor readings, Physiological signals), Temporal granularity (hourly, daily, monthly data). Types: univariate vs. multivariate time series. Overview of common tasks: Forecasting, Classification, Anomaly detection. Overview of time series modelling pipeline: Data collection, preprocessing, modelling, evaluation, deployment.					
Unit 2 (12 hours)	Classical Time Series Methods. Time series vs. cross-sectional data. Components: trend, seasonality, cyclic, noise, Stationarity and differencing, Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF), Time series decomposition, Power Spectrum. Autoregressive (AR), Moving Average (MA), Autoregressive Moving Average (ARMA), Autoregressive Integrated Moving Average (ARIMA), Seasonal ARIMA (SARIMA), Exponential Smoothing (Single, Holt’s Linear, Holt-Winters Seasonal), Model selection: Akaike Information Criterion (AIC), Bayesian Information Criterion (BIC), cross-validation Forecasting and residual diagnostics.					
Unit 3 (10 Hours)	Data Preparation and Feature Engineering, handling missing time series data, Temporal resampling and windowing, Lag features and rolling statistics, Time-based train/test split, Feature importance in time series.					
Unit 4 (10 Hours)	Machine Learning for Time Series, Regression-based forecasting, Tree-based models: Random Forest (RF), eXtreme Gradient Boosting (XGBoost), Light Gradient Boosting Machine (LightGBM), Feature engineering pipelines for ML, Use of Automated Machine Learning (AutoML) and hyperparameter tuning libraries such as Optuna and Tree-based Pipeline Optimization Tool (TPOT), Evaluation metrics: Mean Absolute Percentage Error (MAPE), Root Mean Squared Error (RMSE), Symmetric Mean Absolute Percentage Error (SMAPE)					

Hands-On Learning and Integrated lab: Each module includes hands-on labs with real datasets: Air Quality Index (AQI), Electricity consumption data, ECG time series (MIT-BIH dataset), Stock prices via Yahoo Finance Application Programming Interface (API), Retail demand forecasting

References

1. Brockwell PJ, Davis RA. *Introduction to Time Series and Forecasting*. Springer International Publishing, 2013.
2. Shmueli G. *Practical Time Series Forecasting: A Hands-On Guide*. Axelrod Schnall Publishers, 2016.
3. Brownlee J. *Introduction to Time Series Forecasting with Python*. Jason Brownlee. 2017
4. Lazzeri F. *Machine Learning for Time Series Forecasting with Python*. Wiley. 2020
5. Chaudhuri TD. *Analysis and Forecasting of Financial Time Series Using R: Models and Applications*. Scholars' Press, 2017.
6. Research Articles from the related popular journals

Course Code		AI-511				
Subject Name		AI Survey				
Contact hours/Credit Scheme (L-T-P-C)		1-0-0-0				
Pre-requisites		None				
Course Objectives						
1. Promote the idea of literature survey in the field of artificial intelligence						
Course Outcomes						
CO1. Abreast with the current state-of-art in the field of artificial intelligence applied to various fields of science, finance and management.						
Mapping						
	PO1	PO2	PO3	PO4	PSO1	PSO2
CO1	✓		✓	✓		✓
Course Contents						
	Literature Survey: students are to be directed towards the practice of reading research papers from reputed peer reviewed journals. Area identification: From the vast space of AI, the students are encouraged to narrow the field of scope to a few AI solution. Shortlisting of applications: Depending upon the area the students are encouraged and mentored to identifying a real-life problem for which the AI solution is needed and does not exist and if it exists then how a better AI solution can be used for value addition. The evaluation of the students will be done on the basis of reposts submitted on a bi-monthly basis about the research papers read and on the basis of a final presentation about the AI solution field and justification for the need of the AI solution.					
References						
1. Research papers from IEEE, ELSIVER, SPINGER, other journals of repute.						

Course Code	AI-513					
Subject Name	Industry/Expert Lecture-I					
Contact hours/Credit Scheme (L-T-P-C)	3-0-0-3					
Pre-requisites	None					
Course Objectives						
1. Enhance the state-of-the-art knowledge of the students in the domain of Artificial Intelligence.						
2. Giving the industrial exposure of students						
Course Outcomes						
CO1. Able to have collaborations with the AI experts.						
Mapping						
	PO1	PO2	PO3	PO4	PSO1	PSO2
CO1	✓	✓		✓	✓	✓
Course Contents						
	The AI experts would be invited to deliver the expert lectures. The frequency of the lecture would be weekly. The students will have to submit the report corresponding to each expert lecture.					
	The report should include the following points with regards to the expert lecture:					
	1. Title of the lecture					
	2. Information about the expert including name, institute, and research interests					
	3. Domain and subdomain of the topic					
	4. Significance of the topic					
	5. Concepts					
	6. Information gathered about the topic					
	7. Questions that you asked in the session					
	8. Critical analysis					
	9. Take away points					
	10. Summary of the session					
References						
1. Research papers from IEEE, ELSIVER, SPINGER, other journals of repute.						

Course Code	AI 523					
Subject Name	Python Programming Laboratory					
Contact hours/Credit Scheme (L-T-P-C)	0-0-3-2					
Pre-requisites	None					
Course Objectives						
1. Help Students analyze real life situational problems and think creatively about solutions to them. 2. Apply a solution clearly and accurately in a program using Python. 3. Apply their logic to solve and code various data visualizations problems and machine learning approach to programs.						
Course Outcomes						
CO1.Develop a Python program to demonstrate basics of variables and operators. CO2.Perform operations on data structures using conditions and loops in Python CO3.Develop functions for a given problem. CO4.Develop programs involving Strings and Lists to solve real life problems						
Mapping						
	PO1	PO2	PO3	PO4	PSO1	PSO2
CO1	✓	✓	✓		✓	✓
CO2	✓	✓	✓		✓	✓
CO3	✓		✓	✓	✓	✓
CO4		✓	✓	✓	✓	✓
Laboratory Experiments						
	1. Install and configure Python IDE. 2. Write a simple Python program to display messages on screen. 3. Write program using operators. 4. Write Python program to demonstrate the use of conditional statements 5. Write Python program to demonstrate the use of looping statement 6. Write a Python program to create, access, update and delete lists. Write a program to develop a real life list. 7. Write a Python program to create, access, update and delete tuples. Write a Python program to create, access, update and delete sets. 8. Write a Python program to create, access, update and delete dictionaries. Write a program for looping through dictionaries. 9. Write a Python program to demonstrate math and string built in functions. Write a Python program to demonstrate functions with arguments and function returning values. 10. Write a Python program to demonstrate use of built in modules and user defined modules.					

	11. Write a Python program to demonstrate use of built in packages and user defined packages. 12. Write a Python program to demonstrate use of method overloading and method overriding. 13. Write a Python program to demonstrate inheritance including simple and multiple inheritance. 14. Write a Python program to handle user defined and default exceptions for a given problem. 15. Write Python programs to develop real life files, read, write, update files including various modes and methods.
Tools/Platforms/Languages etc.	
	1. Online compiler: https://www.programiz.com/python-programming/online-compiler 2. PyCharm 3. Kite 4. Spyder

Course Code	AI 525					
Subject Name	Advanced Data Structures and Algorithms Laboratory					
Contact hours/Credit Scheme (L-T-P-C)	0-0-3-2					
Pre-requisites	None					
Course Objectives						
1. Understand various data representation techniques in the real world. 2. Implement linear and non- linear data structures. 3. Develop real-time applications using suitable data structure. 4. Identify suitable data structure to solve various computing problems						
Course Outcomes						
CO1. Analyse a problem, identify and define computing requirements, design and implement solutions. CO2. Solve complex heterogeneous data intensive analytical based problems of real time scenario using state of the art hardware/software tools. CO3. Independently carry out research/investigation and development work to solve practical problems.						
Mapping						
	PO1	PO2	PO3	PO4	PSO1	PSO2
CO1	✓	✓	✓		✓	✓
CO2	✓	✓	✓		✓	✓
CO3	✓		✓	✓	✓	✓
Laboratory Experiments						
	1. Write Python programs for implementing the following searching techniques. a. Linear search b. Binary search c. Fibonacci search 2. Write Python programs for implementing the following sorting techniques to arrange a list of integers in ascending order. a. Bubble sort b. Insertion sort c. Selection sort d. Quick sort e. Merge sort f. Heap sort 3. Write Python programs to a. Design and implement Stack and its operations using List. b. Design and implement Queue and its operations using List. 4. Write Python programs for the following operations on Single, Double and Circular Linked List. (i) Creation (ii) insertion (iii) deletion (iv) traversal 5. Write Python programs to implement the following traversal algorithms: a. Depth first search. b. Breadth first search c. Binary Search tree					

	6. Write a program to perform the following operations: a) Insert an element into an AVL tree. b) Delete an element from an AVL tree. c) Search for a key element in an AVL tree. 7. Write python program to implement Hashing 8. To create ADT that implements the SET concept. a. Add (new Element) - Place a value into the set b. Remove (element) Remove the value c. Contains (element) Return true if element is in collection d. Size () Return number of values in collection Iterator () Return an iterator used to loop over collection e. Intersection of two sets, f. Union of two sets, g. Difference between two sets, h. Subset 9. Read the marks obtained by students of second year in an online examination of particular subject. Find out maximum and minimum marks obtained in that subject. Use heap data structure. Analyse the algorithm Design a mini project using python which will use the different data structure and show the use of specific data structure on the efficiency (performance) of the code.
Tools/Platforms/Languages etc.	
1. Python IDE/IDLE 2. PyCharm 3. Online Compiler: https://www.online-python.com/online_python_compiler 4. Brython 5. PyJS 6. Nuitka	

Course Code	AI 527					
Subject Name	Machine Learning Laboratory					
Contact hours/Credit Scheme (L-T-P-C)	0-0-3-2					
Pre-requisites	None					
Course Objectives						
1. To develop a strong understanding of core machine learning algorithms by implementing them from scratch. 2. To learn how to apply machine learning algorithms on both synthetic and real datasets. 3. To understand evaluation metrics and performance analysis of machine learning models. 4. To build the ability to integrate multiple ML algorithms for solving practical problems.						
Course Outcomes						
CO1.Implement classical supervised and unsupervised machine learning algorithms from scratch.						
CO2.Apply these algorithms on real-world and synthetic datasets to solve classification, regression, and clustering problems.						
CO3.Analyze and evaluate model performance using appropriate metrics.						
CO4.Understand the mathematical foundations underlying the implemented algorithms.						
Mapping						
	PO1	PO2	PO3	PO4	PSO1	PSO2
CO1	✓	✓			✓	✓
CO2	✓	✓	✓		✓	✓
CO3		✓	✓	✓	✓	✓
CO4		✓		✓	✓	
Laboratory Experiments						
	1. Write Python programs for implementing Linear Regression (OLS) 2. Write Python programs for implementing Logistic Regression 3. Write Python programs to implement k-Nearest Neighbours (KNN) 4. Write Python programs Implement Decision Trees (ID3 or CART) 5. Write Python programs to implement Naive Bayes classifier 6. Write a program Implement Support Vector Machine (linear kernel) 7. Write python program to implement K-Means Clustering 8. Write python program to Implement Principal Component Analysis (PCA) 9. Implement Gradient Descent optimization for any of the above algorithms.					

	10. Apply multiple implemented algorithms on a real dataset, evaluate and compare the performance
Tools/Platforms/Languages etc.	
1. Python 3.x environment (Anaconda recommended) 2. Jupyter Notebook or any IDE (e.g., VS Code, PyCharm) 3. Libraries only for data handling and visualization (e.g., NumPy, Pandas, Matplotlib, Seaborn) 4. Datasets (synthetic datasets generated in code and public datasets like UCI repository or Kaggle)	

Second Semester

Course Code	AI-502					
Subject Name	COMPUTER VISION AND NATURAL LANGUAGE PROCESSING					
Contact hours/Credit Scheme (L-T-P-C)	3-0-0-3					
Pre-requisites	None					
Course Objectives						
1. Provide a strong foundation in computer vision (CV) and natural language processing (NLP), covering key techniques such as image feature extraction, object recognition, text parsing, and semantic analysis. 2. Enable to implement CV and NLP algorithms using Python, OpenCV, TensorFlow, and PyTorch, with applications in image classification, object detection, and text analytics. 3. Familiarize with deep learning models such as CNNs for vision tasks and RNNs, LSTMs, and Transformers for NLP applications, including BERT and GPT architectures. 4. Equip to design and integrate vision and NLP solutions for applications like autonomous systems, medical imaging, sentiment analysis, and speech recognition.						
Course Outcomes						
CO1. Explain fundamental principles of computer vision and natural language processing (NLP), including feature extraction, syntactic analysis, and deep learning models.						
CO2. Apply image processing, object recognition, text processing, and language modeling methods in practical applications.						
CO3. Design and optimize CNNs for vision tasks and RNNs, LSTMs, Transformers for NLP applications.						
CO4. Integrate computer vision and NLP models for applications like image captioning, sentiment analysis, and autonomous systems.						
Mapping						
	PO1	PO2	PO3	PO4	PSO1	PSO2
CO1	✓		✓		✓	✓
CO2	✓		✓		✓	✓
CO3	✓		✓	✓	✓	✓
CO4	✓	✓	✓	✓	✓	✓
Course Contents						
Unit 1 (9 Hours)	Introduction to digital image processing, image representation, and transformations. Image filtering techniques such as Gaussian smoothing, edge detection, and histogram equalization. Feature extraction using SIFT, SURF, and HOG. Object detection techniques including Haar cascades, YOLO, and Faster R-CNN. Fundamentals of image segmentation such as thresholding, clustering-based segmentation (K-means), and region-based methods.					

Unit 2 (7 Hours)	Introduction to deep learning for vision tasks, Convolutional Neural Networks (CNNs), architectures like AlexNet, VGG, ResNet, and EfficientNet. Object tracking methods, including Kalman filters and optical flow. 3D vision concepts such as depth estimation, stereo vision, and structure from motion. Introduction to generative models including GANs and VAEs for image generation and enhancement
Unit 3 (8 Hours)	Norms, Distance Metrics, and Projections. Vector and matrix norms. Euclidean and Mahalanobis distances. Projections onto subspaces and least squares approximation. Calculus for AI: Derivatives and Gradients. Single-variable and multivariable derivatives. Chain rule and gradient vectors. Applications in loss minimization and gradient descent.
Unit 4 (5 Hours)	Text preprocessing techniques including tokenization, stemming, lemmatization and stop-word removal. Basic NLP tasks such as part-of-speech (POS) tagging and named entity recognition (NER). Introduction to language modeling, n-grams, hidden Markov models (HMMs), and statistical methods. Word embeddings using Word2Vec, GloVe, and fastText.
Unit 5 (6 Hours)	Recurrent Neural Networks (RNNs), LSTMs, and GRUs for sequential data processing. Introduction to attention mechanisms and Transformer architectures, including BERT, GPT, and T5. Sequence-to-sequence models for machine translation and text summarization. Sentiment analysis and text classification using deep learning techniques.
Unit 6: (5 Hours)	Image captioning using CNN-RNN models, Visual Question Answering (VQA) using attention mechanisms. Implementation of face recognition, object detection and speech-to-text models. Ethical considerations in AI and bias mitigation in language and vision models.
References	
1. Goodfellow I, Bengio Y, Courville A. <i>Deep Learning</i>. MIT Press. 2016. 2. Szeliski R. <i>Computer Vision: Algorithms and Applications</i>. Springer, 2022. 3. Forsyth DA, Ponce J. <i>Computer Vision: A Modern Approach</i>. Pearson, 2011. 4. Jurafsky D, Martin JH. <i>Speech and Language Processing</i>. Pearson, 2021. 5. Manning CD, Schütze H. <i>Foundations of Statistical Natural Language Processing</i>. MIT Press, 1999. 6. Chollet F. <i>Deep Learning with Python</i>. Manning Publications, 2021. 7. Research papers from IEEE and other reputed publishing houses	

Course Code	AI-504					
Subject Name	FOUNDATION OF DEEP LEARNING					
Contact hours/Credit Scheme (L-T-P-C)	3-0-0-3					
Pre-requisites	None					
Course Objectives						
1. To provide a strong foundation in deep learning principles, including neural networks, optimization techniques, and model evaluation. 2. To enable students to develop, train, and fine-tune deep learning models for various applications in computer vision, natural language processing, and generative AI. 3. To impart knowledge on advanced architectures such as CNNs, RNNs, Transformers, and GANs for solving real-world problems. 4. To emphasize the ethical considerations, fairness, and interpretability of AI models in industrial and societal applications.						
Course Outcomes						
CO1. Understand the fundamentals of deep learning, including neural networks, optimization algorithms, and model regularization techniques.						
CO2. Design and implement deep learning architectures such as CNNs and RNNs for solving complex real-world problems in vision and NLP.						
CO3. Utilize generative models like GANs and VAEs for applications in image synthesis, text generation, and reinforcement learning.						
CO4. Analyze and address ethical considerations, fairness, and bias in AI models, ensuring responsible AI deployment.						
Mapping						
	PO1	PO2	PO3	PO4	PSO1	PSO2
CO1	✓		✓		✓	
CO2	✓		✓		✓	✓
CO3	✓	✓	✓		✓	✓
CO4		✓		✓		✓
Course Contents						
Unit 1 (9 Hours)	Introduction to artificial intelligence and machine learning concepts. Fundamentals of neural networks, including perceptrons and multi-layer perceptrons (MLP). Activation functions such as sigmoid, tanh, ReLU, and softmax. Loss functions and optimization techniques including gradient descent and backpropagation. Basics of weight initialization, overfitting, and regularization strategies like dropout and L2 regularization.					
Unit 2 (7 Hours)	Architectures of deep neural networks (DNNs), forward and backward propagation, and computational graphs. Challenges in training deep networks including vanishing and exploding gradients, batch normalization, and parameter tuning techniques. Advanced optimization methods such as Adam, RMSprop, and AdaGrad. Hyperparameter tuning using grid search and random search. Model evaluation metrics including accuracy, precision, recall, and F1-score.					
Unit 3 (8 Hours)	Introduction to convolutional operations, filters, feature maps, and pooling techniques. CNN architectures such as LeNet, AlexNet, VGG, ResNet, and					

	EfficientNet. Transfer learning and fine-tuning pre-trained models. Object detection techniques including Faster R-CNN, YOLO, and SSD. Image segmentation methods such as U-Net and Mask R-CNN. Applications of CNNs in medical imaging, autonomous driving, and facial recognition.
Unit 4 (8 Hours)	Introduction to sequential data processing with RNNs. Challenges and vanishing gradients problem while training RNNs and solutions using LSTMs and GRUs. Forecasting and anomaly detection with LSTMs, 1D Convolutional Neural Networks (1D CNNs) for Time Series, Use in classification and segmentation tasks, Attention and Transformers for Time Series, Temporal Fusion Transformer (TFT), Time Series Transformer (TST), Encoder-decoder architectures. Attention mechanisms and Transformer architectures including BERT, GPT, and T5. Sequence-to-sequence models for applications in machine translation, text generation, and speech recognition. Implementation of Named Entity Recognition (NER), sentiment analysis, and chatbot development using deep learning techniques.
Unit 5 (8 Hours)	Generative models including Variational Autoencoders (VAEs) and Generative Adversarial Networks (GANs). Reinforcement learning fundamentals and applications in AI-driven decision-making. Graph Neural Networks (GNNs) and their applications in social network analysis, chemistry, and recommendation systems. Ethical considerations in AI, fairness, bias mitigation, and explainability in deep learning models.
References	
1. Goodfellow I, Bengio Y, Courville A. <i>Deep Learning</i>. MIT Press. 2016. 2. Chollet F. <i>Deep Learning with Python</i>. Manning Publications, 2021. 3. Bishop C. <i>Pattern Recognition and Machine Learning</i>. Springer, 2006. 4. Haykin S. <i>Neural Networks and Learning Machines</i>. Pearson, 2009. 5. Nielsen M. <i>Neural Networks and Deep Learning</i>. Determination Press, 2015. 6. Géron A. <i>Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow</i>. O'Reilly, 2022. 7. Research papers from IEEE and other reputed publishing houses	

Course Code	AI-506					
Subject Name	Concurrent Programming					
Contact hours/Credit Scheme (L-T-P-C)	3-0-0-3					
Pre-requisites	None					
Course Objectives						
1. Understand the fundamentals of parallel and concurrent programming, including shared vs. distributed memory models. 2. Gain in-depth knowledge of Graphics Processing Unit (GPU) architectures and the CUDA programming model. 3. Apply synchronization techniques to avoid concurrency issues like race conditions and deadlocks. 4. Develop hands-on experience through real-world projects involving scientific computing, image processing, or machine learning acceleration.						
Course Outcomes						
CO1. Differentiate between types of parallelism and concurrency models, and classify systems using Flynn’s taxonomy. CO2. Describe and analyze GPU architecture components such as Streaming Multiprocessors, warps, blocks, and memory hierarchy. CO3. Write, compile, and execute efficient parallel programs using the CUDA programming model. CO4. Apply thread synchronization, atomic operations, and inter-thread communication techniques to ensure correctness and performance.						
Mapping						
	PO1	PO2	PO3	PO4	PSO1	PSO2
CO1			✓		✓	✓
CO2		✓		✓	✓	✓
CO3	✓	✓			✓	
CO4	✓		✓	✓	✓	✓
Course Contents						
Unit 1 (6 Hours)	Introduction to Parallel and Concurrent Programming, Motivation for parallel computing, Types of parallelism: data vs task parallelism, Flynn's taxonomy, Shared memory vs distributed memory models, Introduction to concurrency concepts.					
Unit 2 (7 Hours)	GPU Hardware Architecture, Overview of GPU vs CPU architectures, Streaming Multiprocessors (SMs), Threads, warps, blocks, grids, Memory hierarchy: global, shared, local, texture, constant memory, GPU interconnects and data transfer bottlenecks.					
Unit 3 (9 Hours)	GPU Programming with CUDA, CUDA programming model and syntax, Thread organization: blocks and grids, Memory management in CUDA, launching kernels and basic CUDA API usage, Debugging CUDA programs.					

Unit 4 (9 Hours)	Synchronization and Communication, Thread synchronization within blocks, Race conditions and atomic operations, Memory fences and barriers, Inter-block communication techniques and workarounds, Performance Optimization, Occupancy and utilization, Warp divergence and branch optimization, Coalesced memory access and shared memory usage, Instruction-level and memory-level parallelism, Using CUDA Profiler and Nsight for analysis.
Unit 5 (9 Hours)	Advanced CUDA Features, Streams and asynchronous execution, CUDA events and timing, Unified memory and memory transfers, Dynamic parallelism, Alternative Programming Models, OpenCL basics and comparison with CUDA, Thrust and cuBLAS libraries, OpenACC directives, Introduction to SYCL and oneAPI
Hands-On Learning and Integrated lab: Weekly programming assignments, CUDA kernel writing and debugging, Optimization challenges, Profiling exercises using Nsight/CUDA Profiler, Capstone GPU-accelerated application	
References	
1. Sanders J, Kandrot E. <i>CUDA by Example</i>. Pearson Education, 2010. 2. Kurk DB, Hwu WW. <i>Programming Massively Parallel Processors, A Hands-on Approach</i>. Morgan Kaufmann, 2010 3. Shane C. <i>CUDA Programming</i>. Elsevier Science. 2012 4. NVIDIA CUDA Toolkit Documentation 5. CUDA samples and SDKs from NVIDIA 6. Research Articles from the related popular journals	

Course Code	AI-508					
Subject Name	Reinforcement Learning					
Contact hours/Credit Scheme (L-T-P-C)	3-0-0-3					
Pre-requisites	None					
Course Objectives						
1. To understand the fundamental theory of Reinforcement Learning and its relationship to broader machine learning paradigms including supervised, unsupervised, and deep learning.						
2. To develop the ability to model real-world sequential decision-making problems as Markov Decision Processes and identify suitable RL frameworks for solving them.						
3. To distinguish between model-based and model-free reinforcement learning approaches and apply appropriate methods based on the problem characteristics.						
4. To formulate, implement, and evaluate RL algorithms through computational experiments, with a focus on exploration-exploitation trade-offs and performance analysis for different applications						
Course Outcomes						
CO1. Formulate and solve decision-making problems using RL frameworks.						
CO2. Implement model-free and model-based RL algorithms.						
CO3. Apply function approximation methods using neural networks to enable scalable RL solutions.						
CO4. Design and analyze experiments for RL agents in simulated or real environments.						
Mapping						
	PO1	PO2	PO3	PO4	PSO1	PSO2
CO1	✓		✓	✓	✓	✓
CO2	✓	✓			✓	✓
CO3	✓		✓		✓	✓
CO4	✓	✓	✓	✓	✓	✓
Course Contents						
Unit 1 (3 Hours)	Introduction: What is Reinforcement Learning? Elements of Reinforcement Learning, Comparison with Supervised and Unsupervised Learning, Applications of Reinforcement Learning in different domains: games, robotics, healthcare, finance etc.					
Unit 2 (4 Hours)	Multi-armed Bandits: Exploration vs Exploitation, ϵ -Greedy Strategy, Upper Confidence Bound action selection, Gradient Bandit Algorithm, Contextual Bandits					
Unit 3 (6 Hours)	Markov Decision Processes: Agent-Environment Interface, Goals and Rewards, Returns and Episodes, Policies and Value Functions, Optimality and Approximation					
Unit 4 (7 Hours)	Dynamic Programming: Policy Evaluation, Policy Improvement, Policy and Value Iteration, Asynchronous Dynamic Programming, Efficiency of Dynamic Programming					
Unit 5 (6 Hours)	Model-Free Prediction and Control: Monte Carlo Methods, Temporal Difference Learning, Sarsa and Q-Learning, Eligibility Traces and TD(λ)					
Unit 6 (5 Hours)	Approximation Solution Methods: On-policy Prediction and Control with Approximation, Off-policy methods with Approximation					

Unit 7 (5 Hours)	Deep Reinforcement Learning: Deep neural network based methods for both value-based and policy-based reinforcement learning algorithms
Unit 8 (4 Hours)	Applications and Case Studies
This course will follow an integrated lab model, wherein theoretical instruction is closely coupled with hands-on lab components. This approach ensures experiential learning, enabling students to apply AI concepts through practical exercises, coding assignments, and real-world datasets.	
References	
1. Sutton R, Barto A. <i>Reinforcement Learning: An Introduction</i>. MIT Press, 2nd Edition, 2018. 2. Szepesvari C. <i>Algorithms for Reinforcement Learning</i>. Morgan and Claypool Publishers, 2010 3. Wiering M, Otterlo MV. <i>Reinforcement Learning: State-of-the-Art</i>, Springer, 2012. 4. Bilgin E. <i>Mastering Reinforcement Learning with Python</i>, Packt Publishing, 2020. 5. Lapan M. <i>Deep Reinforcement Learning Hands-On</i>, Packt Publishing, Third Edition, 2024. 6. Research Articles from the related popular journals	

Course Code	AI-510					
Subject Name	Industry/Expert Lecture-II					
Contact hours/Credit Scheme (L-T-P-C)	1-0-0-0					
Pre-requisites	None					
Course Objectives						
1. Enhance the state-of-the-art knowledge of the students in the domain of Artificial Intelligence.						
2. Giving the industrial exposure of students						
Course Outcomes						
CO1. Able to have collaborations with the AI experts.						
Mapping						
	PO1	PO2	PO3	PO4	PSO1	PSO2
CO1	✓	✓		✓	✓	✓
Course Contents						
	The AI experts would be invited to deliver the expert lectures. The frequency of the lecture would be weekly. The students will have to submit the report corresponding to each expert lecture. The report should include the following points with regards to the expert lecture: 1. Title of the lecture 2. Information about the expert including name, institute, and research interests 3. Domain and subdomain of the topic 4. Significance of the topic 5. Concepts 6. Information gathered about the topic 7. Questions that you asked in the session 8. Critical analysis 9. Take away points 10. Summary of the session					
References						
1. Research papers from IEEE, ELSIVER, SPINGER, other journals of repute.						

Course Code	AI-512					
Subject Name	AI Design					
Contact hours/Credit Scheme (L-T-P-C)	1-0-0-0					
Pre-requisites	None					
Course Objectives						
1. Train the students for the thought process which goes into designing and executing an idea based on an AI solution.						
Course Outcomes						
CO1. Understand the application of AI knowledge to solve a real-life problem.						
CO2. Execute training and testing of a scaled AI solution.						
CO3. Deploy the developed application to perform a pilot study of an AI solution.						
Mapping						
	PO1	PO2	PO3	PO4	PSO1	PSO2
CO1	✓	✓	✓	✓	✓	✓
CO2		✓	✓		✓	✓
CO3	✓	✓			✓	
Course Contents						
	Conceptualization of the problem: Introduction to field of application, the current problem and the literature review of the already existing solutions. Framework Design: design of the general framework of the solution, drawing detailed schematics or pipeline diagrams. AI solution shortlisting: classification or regression category identification, finding the best-suited solutions and optimization of the basic design. Data Set identification/collection: Collection of data, identification of data set available from literature. Datamining. Training and Testing: undertake training of the solution (supervised or unsupervised), testing and calculation of performance indexes for the AI designed. The evaluation of the students will be done on bi-monthly presentations and a final report along with presentation, which would give the details of the AI architecture proposed for the AI solution.					
References						
1. Research papers from IEEE, ELSIVER, SPINGER, other journals of repute.						

Course Code	AI-514					
Subject Name	Artificial Intelligence Governance					
Contact hours/Credit Scheme (L-T-P-C)	1-0-0-0					
Pre-requisites	None					
Course Objectives						
1. Help students distinguish between values and skills, and understand the need, basic guidelines, content and process of value education.						
2. Help students initiate a process of dialog within themselves to know what they ‘really want to be’ in their life and profession.						
3. Help students understand the meaning of happiness and prosperity for a human being.						
Course Outcomes						
CO1. Understand the significance of value inputs in a classroom, distinguish between values and skills, understand the need, basic guidelines, content and process of value education,						
CO2.Distinguish between the Self and the Body, understand the meaning of Harmony in the Self the Co-existence of Self and Body.						
CO3.Understand the value of harmonious relationship based on trust, respect and other naturally acceptable feelings.						
Mapping						
	PO1	PO2	PO3	PO4	PSO1	PSO2
CO1	✓	✓	✓		✓	✓
CO2		✓	✓	✓		✓
CO3		✓		✓		✓
Course Contents						
Unit 1	AI Ethics Foundations. Historical context: From Asimov to algorithmic bias. Key principles: fairness, accountability, transparency, privacy (FAT-P). Bias in datasets and model outcomes. Real-world ethical failures. Ethical AI lifecycle					
Unit 2	AI Governance and Responsible AI. What is AI governance? Stakeholders: regulators, developers, industry, civil society. Frameworks: EU AI Act, OECD AI Principles, NIST AI Risk Management Framework, Model auditing, documentation (Model Cards, Datasheets for Datasets), AI governance in organizations: roles, responsibilities, and tools					
Unit 3	AI Containerization & Reproducibility. Introduction to containerization. Virtual machines vs containers. Benefits of containers in AI/ML. Docker Essentials for AI: Dockerfiles for ML models, Volume management and environment control, Managing dependencies across AI workflows. Kubernetes Basics: Pods, nodes, services, Autoscaling and managing AI workloads, Deploying scalable inference services. Reproducibility Tools: MLflow, DVC (Data Version Control).					
Unit 4	Explainable AI (XAI) Techniques, What is XAI and why it matters: Regulatory drivers (e.g., GDPR’s “right to explanation”), Stakeholder trust					

	and model acceptance. Global vs Local Interpretability: Transparent models (decision trees, linear regression), Post-hoc methods for deep learning and ensembles. Key Techniques: LIME (Local Interpretable Model-agnostic Explanations), SHAP (SHapley Additive exPlanations), Counterfactual explanations, Integrated Gradients (for deep learning), Grad-CAM (for image models). Tools: Explainable AI libraries in Python (SHAP, LIME, Captum, Alibi, etc.). Explainability dashboards (What-If Tool, IBM AI Explainability 360)
Unit 5	Case Studies and Capstone Projects: Case 1: Facial recognition ethics & deployment. Case 2: Explainable models in medical AI. Case 3: Containerized XAI-based decision system for loan processing. Capstone: Students build a containerized AI model with explainability dashboard and ethical compliance documentation.
References	
1. Dignum V. Responsible Artificial Intelligence: <i>How to Develop and Use AI in a Responsible Way</i>. Springer International Publishing, 2019 2. Stahl BC. <i>Artificial Intelligence for a Better Future. An Ecosystem Perspective on the Ethics of AI and Emerging Digital Technologies</i>. Springer International Publishing, 2021 3. Muller VC. <i>Risks of Artificial Intelligence</i>. CRC Press. 2016 4. Grant T, Karau H, Lublinsky B, Liu R. <i>Kubeflow for Machine Learning</i>. O'Reilly Media. 2020 5. Research Articles from the related popular journals	

Course Code	AI 524					
Subject Name	Deep Learning Laboratory					
Contact hours/Credit Scheme (L-T-P-C)	0-0-3-2					
Pre-requisites	None					
Course Objectives						
1. Impart practical knowledge on Deep Learning algorithms and their application in different fields.						
Course Outcomes						
CO1.Carry out fundamental and applied research using deep learning mechanisms.						
CO2.Identify innovative research directions in Artificial Intelligence, Machine Learning and Deep Learning.						
CO3.Acquire skills in Deep Learning domain leading to quality publications and innovative models.						
Mapping						
	PO1	PO2	PO3	PO4	PSO1	PSO2
CO1	✓	✓	✓		✓	✓
CO2	✓	✓	✓		✓	✓
CO3	✓		✓	✓	✓	✓
Laboratory Experiments						
	1. Introduction to existing DL platforms and their set-up.					
	2. Introduction to popular DL libraries.					
	3. Development of DL model for forecasting by employing Titanic Survival dataset.					
	4. Development and analysis of custom and benchmark datasets for object detection and tracking.					
	5. Train a Deep learning model to classify a given image using pre-trained model.					
	6. Object detection using Convolution Neural Network.					
	7. Single and multiple object tracking using DL models.					
	8. Recommendation system from sales data using Deep Learning.					
	9. Performance comparison of YOLO, Faster RCNN and SSD for satellite/aerial imagery.					
	10.Hyper parameters tuning to improve the performance of Deep learning models.					
	11.Perform Sentiment Analysis in network graph using RNN.					
	12.Image generation using GAN.					
Tools/Platforms/Languages etc.						
1. Python IDE/IDLE						

2. PyCharm
3. Online Compiler: https://www.online-python.com/online_python_compiler
4. Brython
5. Keras, Pytorch

Course Code	AI 526					
Subject Name	Concurrent Programming Laboratory					
Contact hours/Credit Scheme (L-T-P-C)	0-0-3-2					
Pre-requisites	None					
Course Objectives						
1. To introduce students to GPU architecture and parallel programming concepts.						
2. To teach basic CUDA programming skills including kernel design and memory management.						
3. To enable students to write parallel algorithms exploiting GPU hardware.						
4. To develop understanding of thread hierarchy and memory hierarchy in CUDA						
Course Outcomes						
CO1.Understand and explain the architecture of NVIDIA GPUs and CUDA programming model.						
CO2.Write CUDA kernels to implement parallel algorithms using threads, blocks, and grids.						
CO3.Manage different types of GPU memory (global, shared, local) effectively.						
CO4.Implement and debug parallel algorithms such as vector addition, matrix multiplication, and reduction.						
Mapping						
	PO1	PO2	PO3	PO4	PSO1	PSO2
CO1	✓	✓	✓		✓	
CO2	✓	✓				✓
CO3	✓		✓	✓	✓	✓
CO4	✓	✓		✓	✓	
Laboratory Experiments						
	1. Introduction to CUDA, environment setup, basic kernel launch.					
	2. Memory Management: CUDA memory hierarchy: global, shared, local memory					
	3. Implementation of Thread hierarchy: threads, blocks, grids					
	4. Basic Kernel Programming: Vector addition using CUDA					
	5. Advanced Kernel Programming: Parallel reduction (sum of array elements)					
	6. Matrix multiplication basic implementation					
	7. Parallel patterns: Parallel prefix sum (scan)					
	8. Simple image processing task (e.g., grayscale conversion).					
	9.. Implement Histogram calculation in CUDA.					
	10. Implement basic sorting algorithm (e.g., bitonic sort) in CUDA					
Tools/Platforms/Languages etc.						

1. CUDA Toolkit (compatible with your OS and GPU)
2. NVIDIA GPU-enabled system or CUDA-compatible cloud GPU instance
3. CUDA-capable IDE or editor (e.g., Nsight Eclipse Edition, Visual Studio with CUDA plugin, or VS Code)
4. Sample datasets (synthetic data generated within code and real datasets like image files)
5. Basic C/C++ compiler as CUDA is an extension of C++

Specialization Bouquet 1: Cloud Computing

Course Code	AI 601					
Subject Name	Cloud Infrastructure for AI					
Contact hours/Credit Scheme (L-T-P-C)	3-0-0-3					
Pre-requisites	None					
Course Overview						
This course provides a foundational understanding of the intertwined fields of Data Science and Artificial Intelligence. It explores how data-driven insights are crucial for building intelligent systems and how AI techniques enable sophisticated analysis and decision-making from vast datasets. Students will learn the end-to-end data science pipeline, from data acquisition and preprocessing to exploratory analysis, model development using machine learning algorithms, and the deployment and interpretation of AI solutions.						
Course Objectives						
1. Comprehend the complete data science pipeline, from data collection and cleaning to analysis, modeling, and deployment. 2. Utilize fundamental statistical and visual techniques for exploratory data analysis (EDA) to derive insights and prepare data for AI models. 3. Apply and implement core supervised and unsupervised machine learning algorithms to solve data-driven prediction and pattern recognition problems. 4. Assess the performance of AI models using appropriate metrics.						
Course Outcomes						
CO1. Perform essential data preprocessing and exploratory data analysis (EDA) to prepare datasets for AI and machine learning tasks. CO2. Select and apply appropriate machine learning algorithms (e.g., regression, classification, clustering) to various datasets for specific problem types. CO3. Evaluate the performance of machine learning and AI models using standard metrics and diagnose common issues like overfitting and underfitting. CO4. Discuss the foundational role of data science in developing AI systems and identify key ethical considerations in the use of data and AI technologies.						
Mapping						
	PO1	PO2	PO3	PO4	PSO1	PSO2
CO1					✓	
CO2			✓	✓	✓	✓
CO3	✓				✓	✓
CO4	✓	✓	✓	✓		✓
Course Contents						
Unit 1	Foundations of Data Science and AI: Introduction to Data Science: What is Data Science? The Data Science Lifecycle: Problem understanding, data collection, cleaning, analysis, modeling, deployment. Overview of AI subfields: ML, Deep Learning, NLP, CV. Synergy between Data Science and AI: Data as fuel for AI. Data Types and Sources: Structured vs. Unstructured Data. Tools and Environments: Overview of Python for Data Science (NumPy, Pandas, Matplotlib, Seaborn).					
Unit 2	Data Preprocessing and Exploratory Data Analysis (EDA): Data Loading and Inspection: various data formats (CSV, Excel). Data Cleaning and Preprocessing: Handling missing values (imputation, removal). Dealing with outliers (detection and treatment). Data transformation (scaling, normalization, standardization). Exploratory Data Analysis (EDA): Univariate Analysis: Histograms, box plots, descriptive statistics. Bivariate Analysis: Scatter plots, correlation matrices, cross-tabulations. Multivariate Analysis. Data visualization principles: Choosing appropriate plots.					

Unit 3	Foundational Machine Learning Algorithms: Supervised vs. Unsupervised Learning. Training, Validation, and Test Sets. Overfitting and Underfitting. Regression: Conceptual understanding of Linear Regression. Polynomial Regression (basic idea). Evaluation metrics: MSE, R-squared. Classification: Conceptual understanding of Logistic Regression. Decision Trees and Random Forests. Support Vector Machines (SVMs). Evaluation metrics: Accuracy, Precision, Recall, F1-Score, Confusion Matrix. Unsupervised Learning: Clustering: K-Means algorithm. Dimensionality Reduction: Principal Component Analysis (PCA). Deep Learning for Data Analytics.
Unit 4	Model Evaluation, Deployment, and Responsible AI: Advanced Model Evaluation: Cross-validation techniques (K-Fold). Hyperparameter tuning (Grid Search, Random Search). Model Interpretability: Why interpretability matters for trust and debugging. Basic ideas of feature importance. Introduction to Model Deployment: What it means to deploy an AI model. Ethics in Data Science and AI: Bias and fairness in data and algorithms. Privacy and data security concerns. Transparency and accountability. Responsible AI development principles.
References	
1. Kleppmann M. Designing Data-Intensive Applications. O'Reilly Media, 1st Edition, 2017. 2. Karamitsos I, Graf J. Architecting Data and Machine Learning Platforms: A Guide to Unlocking the Full Potential of Your Data. O'Reilly Media, 1st Edition, 2022. 3. Negnevitsky M. Artificial Intelligence: A Guide to Intelligent Systems. Addison-Wesley, 3rd Edition, 2011. 4. Liebowitz J. Data Analytics and AI. Routledge, 1st Edition, 2020. 5. Crickard P. Data Engineering with Python: Work with Massive Datasets, Manage Data Pipelines, and Build AI Applications. Packt Publishing, 1st Edition, 2021.	

Course Code	AI-602					
Subject Name	Cloud-Native AI Development					
Contact hours/Credit Scheme (L-T-P-C)	3-0-0-3					
Pre-requisites	None					
Course Overview						
This course explores the development of Artificial Intelligence applications using cloud-native paradigms. It delves into building scalable and resilient AI solutions by leveraging serverless computing, container orchestration, microservices architectures, and event-driven programming on cloud platforms.						
Course Objectives						
1. To understand the principles of cloud-native development and their applicability to AI.						
2. To design and implement AI microservices using serverless and containerization technologies.						
3. To integrate various cloud AI services into cloud-native applications.						
4. To build event-driven AI architectures for real-time processing and responsiveness.						
Course Outcomes						
CO1. Develop and deploy scalable AI applications following cloud-native principles.						
CO2. Break down monolithic AI systems into manageable microservices.						
CO3. Utilize cloud-native AI services to enhance application functionality.						
CO4. Create event-driven AI systems for efficient and responsive data processing.						
Mapping						
	PO1	PO2	PO3	PO4	PSO1	PSO2
CO1	✓		✓		✓	✓
CO2		✓				✓
CO3		✓			✓	
CO4	✓	✓	✓	✓	✓	✓
Course Contents						
Unit 1	Introduction to Cloud-Native AI: What is Cloud-Native? Principles, characteristics (containers, microservices, APIs, DevOps). Benefits of cloud-native approach for AI development. Monolithic vs. Microservices architectures for AI applications. Introduction to serverless and container orchestration in cloud-native AI.					
Unit 2	Containerization and Orchestration for AI: Deep dive into Docker for AI: Image optimization, multi-stage builds. Kubernetes for AI workloads: Pods, deployments, services, ingress. Managed Kubernetes services (EKS, AKS, GKE) for scalable AI deployments. Running distributed AI training and inference on Kubernetes. Helm charts for packaging and deploying AI applications.					
Unit 3	Serverless AI: Serverless computing for AI: When to use, benefits, limitations. AWS Lambda, Azure Functions, Google Cloud Functions for AI inference and data processing. Building serverless API gateways for AI models. Orchestrating serverless functions with Step Functions, Azure Logic Apps, Cloud Workflows. Use cases for serverless AI.					
Unit 4	Integrating Cloud AI Services: Cognitive Services/Pre-trained AI APIs: Vision, Speech, Language, Search (e.g., AWS Rekognition, Azure Cognitive Services, Google Cloud Vision API). Integrating these services into cloud-native AI applications. Managed Machine Learning APIs (e.g., AWS Comprehend, Azure Text Analytics, Google Natural Language API). Building composite AI solutions using multiple cloud services.					
Unit 5	Event-Driven AI Architectures: Introduction to event-driven architectures.					

	Message queues and streaming services: AWS SQS/SNS, Azure Service Bus/Event Hubs, Google Pub/Sub. Building real-time AI pipelines with event sources and serverless functions. Change Data Capture (CDC) for AI data pipelines. Use cases: Real-time fraud detection, recommendation systems, personalized experiences.
Unit 6	Cloud-Native Development Best Practices: Observability for cloud-native AI: Logging, monitoring, tracing (e.g., Prometheus, Grafana, OpenTelemetry). CI/CD for cloud-native AI applications. Security in cloud-native environments. Resiliency patterns for distributed AI systems.
References	
1. Ajit Pratap Kundan (Author), Mainak Chakraborty. <i>Monitoring Cloud-Native Applications: Lead Agile Operations Confidently Using Open Source Software</i>. Academic Press, 2021. 2. David Bounds, Premkumar Rangarajan. <i>Cloud Native AI and Machine Learning on AWS</i>, bpb, 2023. 3. Adrian Gonzalez Sanchez. <i>Azure OpenAI Service for Cloud Native Applications: Designing, Planning, and Implementing Generative AI Solutions</i>. Shroff/O'Reilly, 2024. 4. Avinash Pamisetty. <i>Cloud-Native Intelligence and Scalable AI for Supply Chain, Food Services, and Financial Infrastructure</i>, Aqua Publications, 2025. 5. Online Resources from various Cloud Service Providers	

Course Code	AI-603					
Subject Name	MLOps on Cloud Platforms					
Contact hours/Credit Scheme (L-T-P-C)	3-0-0-3					
Pre-requisites	None					
Course Overview						
This course focuses on the principles and practices of MLOps (Machine Learning Operations) within cloud environments. Students will learn how to build, deploy, and manage the entire machine learning lifecycle, from data preparation and model training to deployment, monitoring, and retraining leveraging cloud-native tools and services.						
Course Objectives						
1. To understand the end-to-end MLOps lifecycle and its importance in cloud AI development. 2. To leverage cloud services for efficient data versioning, feature engineering, and data pipeline orchestration. 3. To implement automated model training, evaluation, and versioning workflows on cloud platforms. 4. To deploy, monitor, and manage machine learning models in production environments on the cloud.						
Course Outcomes						
CO1. Design and implement MLOps pipelines for machine learning projects on cloud platforms. CO2. Utilize cloud services for automated data processing and feature store management. CO3. Automate model training and evaluation using cloud-based ML services. CO4. Deploy and monitor ML models effectively in cloud production environments, ensuring performance and reliability.						
Mapping						
	PO1	PO2	PO3	PO4	PSO1	PSO2
CO1		✓			✓	✓
CO2		✓		✓	✓	✓
CO3	✓			✓	✓	✓
CO4	✓	✓	✓	✓	✓	✓
Course Contents						
Unit 1	Introduction to MLOps: What is MLOps? Principles, benefits, and challenges. The MLOps lifecycle: Data preparation, model development, deployment, monitoring, retraining. Differences between traditional software DevOps and MLOps. Role of cloud platforms in enabling MLOps.					
Unit 2	Data Management and Feature Stores on Cloud: Data versioning for ML datasets (e.g., DVC, cloud object storage versioning). Data pipelines and orchestration: Apache Airflow (managed services like MWAA, Cloud Composer), Azure Data Factory, AWS Step Functions. Feature engineering at scale on the cloud. Introduction to Feature Stores: Concepts, benefits, and cloud-based implementations.					
Unit 3	Cloud-based Model Training and Experiment Tracking: Managed ML services for training: AWS SageMaker, Azure Machine Learning, Google Cloud AI Platform. Distributed training on the cloud (e.g., TensorFlow Distributed, PyTorch Distributed). Experiment tracking and model versioning: MLflow, Azure ML tracking, SageMaker Experiments. Hyperparameter tuning services (e.g., SageMaker Automatic Model Tuning, Azure ML hyperdrive). Model evaluation and validation strategies on cloud.					
Unit 4	Model Deployment and Serving: Batch inference vs. real-time inference. Containerization for model serving (Docker, Kubernetes on EKS/AKS/GKE). Serverless inference (e.g., Lambda, Azure Functions, Cloud Functions for small					

	models). Managed model endpoints: SageMaker Endpoints, Azure ML Endpoints, A/B testing and canary deployments for ML models.
Unit 5	Model Monitoring and Retraining: Monitoring model performance: drift detection (data drift, concept drift), anomaly detection. Monitoring infrastructure metrics (CPU, memory, GPU utilization). Alerting and logging for ML models. Automated model retraining pipelines based on monitoring triggers. Model governance and explainability (XAI) in production.
Unit 6	MLOps Best Practices and Case Studies: Building robust and scalable MLOps pipelines. Security considerations in MLOps. Cost optimization for MLOps workflows. Real-world MLOps case studies on various cloud platforms.
References	
1. Mark Treveil, Nicolas Omont, Clément Stenac. <i>Introducing MLOps: How to Scale Machine Learning in the Enterprise</i>. Shroff/O'Reilly, 2020. 2. Noah Gift, Alfredo Deza. <i>Practical MLOps: Operationalizing Machine Learning Models</i>, Shroff/O'Reilly, 2021. 3. Emmanuel Raj. <i>Engineering MLOps</i>. Packt Publishing, 2021. 4. Saurabh D. Dorle. <i>Ultimate MLOps for Machine Learning Models: Use Real Case Studies to Efficiently Build, Deploy, and Scale Machine Learning Pipelines with MLOps</i>. Kindle Edition, 2024. 5. Yaron Haviv, Noah Gift. <i>Implementing MLOps in the Enterprise: A Production-First Approach</i>. Shroff/O'Reilly, 2023	

Course Code	AI-604					
Subject Name	Advanced Topics in Cloud AI and Responsible AI					
Contact hours/Credit Scheme (L-T-P-C)	3-0-0-3					
Pre-requisites	None					
Course Overview						
This course delves into cutting-edge topics in cloud-based Artificial Intelligence, including specialized cloud ML hardware, federated learning, ethical considerations, and responsible AI practices. Students will explore advanced optimization techniques, privacy-preserving AI, and the societal impact of AI deployed on cloud platforms.						
Course Objectives						
1. To explore advanced cloud infrastructure and services for specialized AI workloads (e.g., large language models, reinforcement learning).						
2. To understand the principles and practical applications of privacy-preserving AI techniques like federated learning on cloud.						
3. To analyze and address ethical considerations and biases in AI systems deployed on cloud platforms.						
4. To learn about responsible AI frameworks, tools, and best practices for developing and deploying fair, accountable, and transparent AI.						
Course Outcomes						
CO1. Design and implement AI solutions leveraging advanced cloud hardware and specialized services.						
CO2. Apply privacy-preserving AI techniques in cloud environments for sensitive data.						
CO3. Identify and mitigate ethical biases in cloud-based AI systems.						
CO4. Develop and deploy responsible AI solutions adhering to fairness, accountability, and transparency principles.						
Mapping						
	PO1	PO2	PO3	PO4	PSO1	PSO2
CO1	✓	✓			✓	
CO2		✓	✓			✓
CO3				✓		✓
CO4	✓	✓		✓	✓	✓
Course Contents						
Unit 1	Advanced Cloud ML Infrastructure and Services: Deep dive into specialized AI accelerators: TPUs (Google Cloud TPU), FPGAs (AWS F1), custom chips. Cloud services for Large Language Models (LLMs) and Generative AI: Fine-tuning, deployment, prompt engineering (e.g., AWS Bedrock, Azure OpenAI Service, Google Cloud Generative AI Studio). Reinforcement Learning platforms on cloud (e.g., AWS SageMaker RL, Azure ML with Ray). High-Performance Computing (HPC) for AI on cloud.					
Unit 2	Privacy-Preserving AI on Cloud: Introduction to privacy concerns in AI: Data privacy, model privacy. Homomorphic Encryption and its application in cloud AI. Differential Privacy: Concepts, implementation, and cloud tooling. Federated Learning: Principles, architectures (e.g., TensorFlow Federated on cloud), use cases (e.g., healthcare, mobile data). Secure Multi-Party Computation (SMC) for collaborative AI.					
Unit 3	Ethical AI and Bias Detection: Introduction to AI ethics: Fairness, accountability, transparency, safety. Sources of bias in AI systems: Data bias, algorithmic bias, societal bias. Techniques for detecting and measuring bias in datasets and models. Mitigation strategies for algorithmic bias. Explainable AI (XAI) in practice: LIME, SHAP, and cloud-based XAI tools.					

Unit 4	Responsible AI Frameworks and Governance: Overview of responsible AI principles and guidelines (e.g., NIST AI Risk Management Framework, EU AI Act). Implementing responsible AI practices throughout the MLOps lifecycle. Model cards and datasheets for transparency. AI governance and regulatory compliance in cloud environments. Building trust in AI systems.
Unit 5	Societal Impact and Future of Cloud AI: The impact of cloud AI on various industries (healthcare, finance, education, transportation). Sustainable AI: Energy consumption and environmental impact of large cloud AI models. The future of cloud AI: Edge AI, hybrid cloud AI, sovereign cloud for AI.
References	
1. Jerry Chou (Editor), Wu-Chun Chung (Editor). <i>Cloud Computing and High Performance Computing</i>. Mdpi AG, 2025. 2. Parth Girish Patel, Ishneet Kaur Dua, Steven David. <i>Sustainable Cloud Development: Optimize cloud workloads for environmental impact in the GenAI era</i>, Packt Publishing, 2025. 3. Denis Rothman. <i>Hands-On Explainable AI (XAI) with Python</i>. Packt Publishing, 2020. 4. Michael Munn, David Pitman. <i>Explainable AI for Practitioners: Designing and Implementing Explainable ML Solutions</i>, Shroff/O'Reilly, 2022. 5. Online Resources from various Cloud Service Providers	

Specialization Bouquet 2: Data Analytics

Course Code	AI-611					
Subject Name	Advanced Data Management, Engineering, and Preprocessing for AI					
Contact hours/Credit Scheme (L-T-P-C)	3-0-0-3					
Pre-requisites	None					
Course Overview						
This course delves into advanced concepts of data acquisition, management, and rigorous preprocessing techniques essential for building robust AI systems. Leveraging students' existing statistical knowledge, it focuses on complex data quality issues, sophisticated imputation methods, comprehensive feature engineering strategies, and the architectural implications of various data storage solutions. The emphasis is on preparing diverse data types for high-performance AI/ML model training.						
Course Objectives						
1. To analyze and design advanced strategies for acquiring, managing, and integrating large-scale, heterogeneous datasets 2. To master sophisticated data cleaning, imputation, transformation, and feature engineering techniques for complex data. 3. To apply in-depth exploratory data analysis and advanced statistical methods to uncover intricate patterns and relationships in data. 4. To evaluate and implement robust data governance principles and ethical considerations in the data pipeline lifecycle.						
Course Outcomes						
Upon completion of this course, students will be able to: CO1. Propose and evaluate architectures for data ingestion and storage across various data paradigms (relational, NoSQL, streaming, data lakes). CO2. Develop and justify comprehensive data preprocessing pipelines that handle real-world data imperfections and prepare data for advanced AI models. CO3. Perform advanced exploratory data analysis, inferring complex data characteristics and preparing insightful visualizations. CO4. Design and advocate for ethical data handling practices and robust data governance frameworks within an AI context.						
Mapping						
	PO1	PO2	PO3	PO4	PSO1	PSO2
CO1	✓		✓		✓	✓
CO2	✓		✓		✓	✓
CO3	✓		✓	✓	✓	✓
CO4	✓	✓	✓	✓	✓	✓
Course Contents						
Unit 1	Advanced Data Acquisition and Storage Architectures. Role of data in AI ecosystems, data lifecycle in complex systems, Advanced techniques for data collection from various sources (APIs, web crawling concepts for large scale, IoT sensors, enterprise systems), Dimensional modeling (Star/Snowflake schema conceptual), ETL/ELT processes, columnar storage concepts, optimization for analytical queries, Conceptual architecture (raw, refined, curated zones), open formats (Parquet, ORC, Avro), schema-on-read vs. schema-on-write. In-depth conceptual understanding of different models (Key-Value, Document, Column-Family, Graph) for specific AI use cases (e.g., social networks for graph DBs, personalized profiles for document DBs), Message queues (Kafka, Kinesis - conceptual roles) for real-time data ingestion, Heterogeneous data sources, schema evolution, data lineage.					

Unit 2	Advanced Data Cleaning and Transformation. Structural errors, semantic inconsistencies, data aging, conceptual drift, data latency, Conceptual understanding of missing data mechanisms (MCAR, MAR, MNAR), Model-based imputation techniques (e.g., K-Nearest Neighbors imputation, Multiple Imputation by Chained Equations - conceptual algorithms), Handling time-series specific missingness, Advanced detection methods (Isolation Forest, One-Class SVM - conceptual understanding); robust statistical methods (e.g., Tukey's fences), Discretization/Binning strategies (equal-width, equal-frequency, custom), Mathematical transformations (log, square root, power transforms - conceptual application for normalization/stabilizing variance), Normalization vs. Standardization (conceptual impact on different AI algorithms), Handling Skewness and Kurtosis (conceptual strategies), Advanced tokenization, n-grams, part-of-speech tagging, named entity recognition (conceptual methods), Normalization, resizing, cropping, color space conversion (conceptual steps).
Unit 3	Feature Engineering and Selection for AI. Translating raw data into impactful features for AI models, domain-driven feature creation, Overview of techniques like featurtools (deep feature synthesis), Target Encoding, Feature Hashing, Embeddings for high-cardinality categorical features (conceptual role), Lag features, rolling statistics (mean, std dev, min, max), time-based features (day of week, month, year, holiday indicators), Fourier transforms for seasonality (conceptual), Node centrality measures, community detection features (conceptual), Mutual Information, Chi-squared (for categorical), ANOVA F-value (for numerical and categorical target), Recursive Feature Elimination (RFE), Sequential Feature Selection (SFS), Feature importance from tree-based models, L1 regularization effects, Conceptual understanding of feature interactions and selection for high-dimensional data.
Unit 4	Advanced Exploratory Data Analysis and Visualization. MANOVA (conceptual), Canonical Correlation Analysis (conceptual) for complex relationships, Autocorrelation Function (ACF), Partial Autocorrelation Function (PACF) plots for identifying patterns and model orders. Decomposition of time series (trend, seasonality, residuals), Choropleth maps, heatmaps for spatial patterns (conceptual), Node-link diagrams, force-directed layouts for relational data (conceptual), Conceptual design of dynamic dashboards, drill-downs, cross-filtering for deeper insights, Advanced conceptual understanding of t-SNE and UMAP for visualizing high-dimensional data clusters and manifolds, Principles of narrative construction, contextualization, and persuasive communication of insights to diverse audiences.
Unit 5	Data Governance and Data Ethics in AI. Data quality management, data lineage, metadata management, data stewardship, data security policies, Advanced anonymization and pseudonymization techniques (k-anonymity, l-diversity, t-closeness - conceptual), Sources of bias (collection bias, historical bias, sampling bias, measurement bias, algorithmic bias), Conceptual methods for bias detection in datasets (e.g., demographic parity, equalized odds - from a data perspective), Strategies for mitigating bias in data (e.g., re-sampling, re-weighting, data augmentation for underrepresented groups - conceptual), Ethical frameworks for data acquisition and usage in AI, In-depth conceptual understanding of global and Indian data protection laws (GDPR, CCPA, Digital Personal Data Protection Act, 2023) and their implications for data handling in AI projects, Importance of tracking data sources and transformations.
<i>This course will follow an integrated lab model, wherein theoretical instruction is closely coupled with hands-on lab components. This approach ensures experiential learning, enabling students to</i>	

apply AI concepts through practical exercises, coding assignments, and real-world datasets.

References

1. Kleppmann M. *Designing Data-Intensive Applications*. O'Reilly Media, 1st Edition, 2017.
2. Karamitsos I, Graf J. *Architecting Data and Machine Learning Platforms: A Guide to Unlocking the Full Potential of Your Data*. O'Reilly Media, 1st Edition, 2022.
3. Negnevitsky M. *Artificial Intelligence: A Guide to Intelligent Systems*. Addison-Wesley, 3rd Edition, 2011.
4. Liebowitz J. *Data Analytics and AI*. Routledge, 1st Edition, 2020.
5. Crickard P. *Data Engineering with Python: Work with Massive Datasets, Manage Data Pipelines, and Build AI Applications*. Packt Publishing, 1st Edition, 2021.

Course Code	AI-612					
Subject Name	Advanced AI for Unstructured Data and Strategic Analytics					
Contact hours/Credit Scheme (L-T-P-C)	3-0-0-3					
Pre-requisites	None					
Course Overview						
This course delves into advanced AI techniques, specifically focusing on deep learning architectures for processing and deriving insights from unstructured data—Natural Language (text) and Vision (images/videos). Students will gain a deep conceptual understanding of modern NLP and Computer Vision models, exploring their application in strategic data analytics, and how these AI models contribute to sophisticated decision support and complex problem-solving.						
Course Objectives						
1. To understand and apply state-of-the-art deep learning architectures for Natural Language Processing (NLP) tasks. 2. To master the conceptual principles of advanced Computer Vision (CV) algorithms for image and video analysis. 3. To design AI-driven analytical solutions that integrate insights from diverse unstructured data sources. 4. To evaluate the performance, biases, and ethical considerations of advanced NLP and CV models in real-world strategic applications						
Course Outcomes						
CO1. Formulate and conceptualize advanced NLP models for tasks such as sentiment analysis, language generation, and information extraction. CO2. Design and explain CV systems for image classification, object detection, and semantic segmentation from visual data. CO3. Propose integrated AI analytics solutions that combine insights from textual and visual data for strategic decision-making. CO4. Critically assess the capabilities, limitations, and potential biases in advanced NLP and CV models for real-world impact.						
Mapping						
	PO1	PO2	PO3	PO4	PSO1	PSO2
CO1	✓		✓		✓	✓
CO2	✓		✓		✓	✓
CO3	✓		✓	✓	✓	✓
CO4	✓	✓	✓	✓	✓	✓
Course Contents						
Unit 1	Deep Learning for Natural Language Processing. Word Embeddings in Depth: Word2Vec (Skip-gram, CBOW conceptual algorithms), GloVe, FastText (conceptual advantages and training), Subword Tokenization: Byte Pair Encoding (BPE), WordPiece (conceptual purpose and benefits), Architectures like Bidirectional LSTMs/GRUs for sequence understanding (e.g., conceptual for Named Entity Recognition, Part-of-Speech Tagging), 1D CNNs for text classification (conceptual filter application on word embeddings),Conceptual understanding of Self-Attention mechanism (Query, Key, Value vectors),Multi-Head Attention, Positional Encoding (conceptual role in sequence order), Encoder-Decoder structure, Conceptual pre-training objectives (Masked Language Model, Next Sentence Prediction) and fine-tuning paradigm for downstream tasks, Conceptual understanding of autoregressive					

	generation, diverse applications (text generation, summarization, dialogue), RoBERTa, XLNet, T5 (conceptual differences and improvements, Strategies for interacting with large language models, Sentiment Analysis and Emotion Detection (fine-grained vs. aspect-based), Topic Modeling (Neural Topic Models - conceptual), Text Summarization (Extractive vs. Abstractive - conceptual approaches), Named Entity Recognition (NER) and Relation Extraction, Question Answering Systems (Conceptual types: Extractive, Generative), Machine Translation (Seq2Seq with attention conceptual), Chatbots and Dialogue Systems (conceptual architectures).
Unit 2	Deep Learning for Computer Vision. Conceptual understanding of hierarchical feature extraction by convolutional layers, Detailed architectural evolution: LeNet, AlexNet, VGG, ResNet (residual connections), Inception (multi-scale processing), DenseNet (conceptual benefits of skip connections), Data Augmentation for Image Data: Conceptual techniques (rotation, scaling, flipping, color jittering) to improve model robustness and generalization, Strategies for fine-tuning pre-trained CNNs (e.g., on ImageNet) for specific image classification tasks; conceptual understanding of feature extractor vs. fine-tuning entire network, Conceptual design of robust image classifiers, Identifying and localizing multiple objects in an image (conceptual algorithms: R-CNN family, YOLO (You Only Look Once), SSD (Single Shot MultiBox Detector) – conceptual principles of bounding boxes and class probabilities), Assigning a class to each pixel (Semantic Segmentation) or identifying individual instances of objects (Instance Segmentation) – conceptual approaches (e.g., U-Net, Mask R-CNN), Combining CNNs and RNNs/Transformers (conceptual architecture), Detailed conceptual understanding of generator-discriminator training, applications in image synthesis, style transfer, super-resolution, Variational Autoencoders (VAEs): Conceptual understanding of latent space and probabilistic generation, Action recognition, activity detection, tracking objects in video.
Unit 3	Multi-Modal AI for Integrated Analytics. Conceptual approaches to combining insights from text, images, tabular data, and audio, When to combine features at different stages of the AI pipeline, Sentiment analysis from text and facial expressions, Product recommendation systems leveraging image, text, and tabular data, Medical diagnosis combining medical images and patient notes, Autonomous driving (conceptual fusion of sensor data: LiDAR, camera, radar), Alignment, representation learning across modalities.
Unit 4	Strategic Analytics and Prescriptive AI Systems. Beyond prediction to recommending optimal actions and achieving desired outcomes, Conceptual framework of Agent-Environment interaction, Reward maximization, Markov Decision Processes (MDPs) and Bellman Equations (conceptual foundations), Value-Based RL (Q-learning, DQN - conceptual), Policy-Based RL (Policy Gradients - conceptual), Applications: Dynamic pricing, inventory optimization, supply chain management, resource allocation, intelligent routing, personalized marketing strategies, Using AI models within simulations for scenario planning and policy evaluation, Linear Programming, Integer Programming, Constraint Programming (brief overview of how AI can solve complex optimization problems), Conceptual design of systems that augment human decision-making with AI insights and recommendations, Understanding cause-and-effect relationships in data, for better decision-making.
Unit 5	Responsible AI and Advanced Interpretability. Fairness, Accountability, Transparency, Robustness, Privacy, Safety, Conceptual understanding of fairness metrics (e.g., demographic parity, equalized odds, counterfactual fairness), Conceptual techniques for debiasing models (pre-processing, in-processing, post-processing methods), LIME and SHAP in detail (conceptual framework, advantages, limitations), Partial Dependence Plots, Individual Conditional Expectation (ICE)

	plots, conceptual sensitivity analysis, Saliency Maps, Class Activation Maps (CAM, Grad-CAM), LRP (Layer-wise Relevance Propagation) – conceptual methods to understand the process of decision making by a deep model, Conceptual frameworks for continuous evaluation, Considering societal impact, unintended consequences, and human-in-the-loop strategies.
<i>This course will follow an integrated lab model, wherein theoretical instruction is closely coupled with hands-on lab components. This approach ensures experiential learning, enabling students to apply AI concepts through practical exercises, coding assignments, and real-world datasets.</i>	
References	
1. Szeliski R. <i>Computer Vision: Algorithms and Applications</i>. Springer, 2nd Edition, 2022. 2. Forsyth DA, Ponce J. <i>Computer Vision: A Modern Approach</i>. Pearson, 2nd Edition, 2011. 3. Jurafsky D, Martin JH. <i>Speech and Language Processing</i>. Pearson, 3rd Edition, 2021. 4. Manning CD, Schütze H. <i>Foundations of Statistical Natural Language Processing</i>, MIT Press, 1st Edition, 1999. 5. Goodfellow I, Bengio Y, Courville A. <i>Deep Learning</i>. MIT Press, 1st Edition, 2016. 6. Chollet F. <i>Deep Learning with Python</i>. Manning Publications, 2nd Edition, 2021.	

Course Code	AI-613					
Subject Name	Advanced Machine Learning Models for Data Analytics					
Contact hours/Credit Scheme (L-T-P-C)	3-0-0-3					
Pre-requisites	None					
Course Overview						
Building on core AI/ML knowledge and advanced data preparation skills, this course dives into sophisticated machine learning models crucial for complex data analytics. It explores advanced ensemble techniques, specialized regression and classification models, advanced clustering methods, and deep learning foundations, emphasizing their theoretical underpinnings, algorithmic nuances, and application in extracting deeper insights and making robust predictions from diverse datasets.						
Course Objectives						
1. To understand and apply advanced ensemble learning methods for superior predictive performance and robustness in analytical tasks. 2. To explore and conceptualize specialized machine learning models for handling complex data distributions and relationships. 3. To master advanced unsupervised learning techniques for uncovering intricate patterns, anomalies, and structures in high-dimensional data. 4. To evaluate the performance, interpretability, and ethical considerations of advanced ML models in real-world data analytics scenarios.						
Course Outcomes						
Upon completion of this course, students will be able to:						
CO1. Design and deploy robust ensemble models to achieve state-of-the-art predictions in challenging data environments.						
CO2.Select and justify the use of specialized machine learning models for non-linear, high-dimensional, and imbalanced data.						
CO3.Discover subtle patterns and outliers in complex datasets using advanced clustering and anomaly detection algorithms.						
CO4.Critically assess the trade-offs between model complexity, performance, and interpretability in advanced analytical applications.						
Mapping						
	PO1	PO2	PO3	PO4	PSO1	PSO2
CO1	✓		✓		✓	✓
CO2	✓		✓		✓	✓
CO3	✓		✓	✓	✓	✓
CO4	✓	✓	✓	✓	✓	✓
Course Contents						
Unit 1	Advanced Ensemble Learning Architectures. Bias-Variance decomposition, stability of ensembles, conceptual role of diversity, Conceptual algorithm, decision boundaries, out-of-bag error, and detailed understanding of feature importance measures, Extra-Trees (Extremely Randomized Trees): Conceptual differences and benefits, Detailed conceptual understanding of additive models, loss functions for different tasks (e.g., LogLoss for classification, Huber loss for robust regression), and iterative residual fitting, Conceptual understanding of regularization (L1/L2), parallel tree boosting, handling missing values, custom objective functions, Conceptual advantages for speed and memory efficiency (Gradient-based One-Side Sampling (GOSS), Exclusive Feature Bundling, Conceptual approach to handling categorical					

	features natively and ordered boosting, Meta-learning techniques, conceptual design of diverse base models and a meta-learner for final prediction; conceptual strategies for preventing information leakage in stacking, When to use which ensemble method, hyperparameter tuning conceptualization for ensembles.
Unit 2	Advanced Regression and Classification Models. Conceptual extension of linear regression for different response distributions (e.g., Logistic Regression as a GLM, Poisson Regression for count data), Maximizing the margin, dual problem conceptualization, Kernel Functions (RBF, Polynomial, Sigmoid - conceptual mechanism to project data into higher dimensions for linear separation), Conceptual understanding of parameters (C, gamma), Conditional Inference Trees, conceptual understanding of impurity measures, stability of trees, Beyond SVMs, general idea of mapping data to feature space for linear separability, Algorithmic approaches (cost-sensitive learning, conceptual understanding of algorithm modifications), Data-level approaches (Oversampling: SMOTE - conceptual algorithm; Undersampling: Tomek Links, NearMiss - conceptual), Conceptual understanding for anomaly detection (e.g., One-Class SVM).
Unit 3	Advanced Unsupervised Learning - Clustering & Anomaly Detection. Conceptual understanding of external validation metrics (e.g., Adjusted Rand Index, Normalized Mutual Information) and internal validation metrics (e.g., Davies-Bouldin Index, Calinski-Harabasz Index), Stability-based clustering evaluation, Conceptual understanding of probabilistic clustering, expectation-maximization (EM) algorithm, Conceptual parameters (epsilon, min_samples), robustness to noise, finding arbitrary shapes, Hierarchical Clustering (Agglomerative, Divisive - detailed conceptual methods and linkage criteria), Spectral Clustering (conceptual understanding based on graph partitioning), Statistical methods (conceptual: Z-score, IQR, Mahalanobis distance), Machine Learning-based methods, Applications in fraud detection, system monitoring.
Unit 4	Deep Learning Foundations for Data Analytics. Multilayer Perceptrons (MLPs) for complex tabular data (conceptual design choices: number of layers, neurons per layer, activation functions), Deep Learning for Feature Learning: Conceptual understanding of how deep networks learn hierarchical representations, Conceptual understanding of momentum, Nesterov momentum, AdaGrad, RMSprop, Adam, learning rate schedules (conceptual benefits of adaptive optimizers), Dropout, Batch Normalization, Conceptual understanding of L1/L2 regularization in deep networks, Reusing pre-trained deep learning models (conceptual benefits for data-scarce domains), Basics of encoding and decoding; use cases for dimensionality reduction, feature learning, and anomaly detection, Basics of generator-discriminator architecture, applications in data augmentation and synthesis.
Unit 5	Time Series Forecasting with Advanced ML. Stationarity, autocorrelation, partial autocorrelation, trend, seasonality, cycles, white noise (conceptual interpretation from ACF/PACF plots), ARIMA, SARIMA for identifying and modeling components, Lag features, rolling statistics, Fourier series terms for seasonality, conceptual understanding of creating calendar-based features, using tree-based models (e.g., XGBoost) on engineered features from time series data, Deep Learning for Time Series and forecasting evaluation.
Unit 6	Model Interpretability and Explainable AI (XAI). Trust, debugging, fairness, regulatory compliance, scientific discovery, Model-agnostic vs. Model-specific, Local vs. Global explanations, Conceptual idea of perturbing inputs and building local interpretable models, Conceptual understanding based on cooperative game theory, attributing feature contributions to individual predictions, Partial Dependence

	Plots (PDP): Conceptual visualization of average marginal effect of a feature, Individual Conditional Expectation (ICE) plots: Conceptual visualization of individual prediction changes, Permutation Importance: Conceptual method for feature importance by shuffling, Conceptual ideas of Saliency Maps, Class Activation Maps (CAM) for understanding important regions in images.
<i>This course will follow an integrated lab model, wherein theoretical instruction is closely coupled with hands-on lab components. This approach ensures experiential learning, enabling students to apply AI concepts through practical exercises, coding assignments, and real-world datasets.</i>	
References	
1. Goodfellow I, Bengio Y, Courville A. <i>Deep Learning</i>. MIT Press, 1st Edition, 2016. 2. Provost F, Fawcett T. <i>Data Science for Business</i>. O'Reilly Media, 1st Edition, 2013. 3. Kleppmann M. <i>Designing Data-Intensive Applications</i>. O'Reilly Media, 1st Edition, 2017. 4. Negnevitsky M. <i>Artificial Intelligence: A Guide to Intelligent Systems</i>. Addison-Wesley, 3rd Edition, 2011. 5. Chollet F. <i>Deep Learning with Python</i>. Manning Publications, 2nd Edition, 2021. 6. Research Articles from the related popular journals	

Course Code	AI-614					
Subject Name	Advanced Data Engineering for AI System					
Contact hours/Credit Scheme (L-T-P-C)	3-0-0-3					
Pre-requisites	None					
Course Overview						
This capstone course integrates all previously learned advanced AI/ML and data science concepts with the practical challenges of deploying and managing AI systems at scale. It covers the architectural design of big data platforms, the comprehensive principles of Machine Learning Operations (MLOps) for the entire AI lifecycle, and the strategic leveraging of cloud-based AI services. The focus is on building, deploying, and maintaining robust, scalable, and ethical AI-driven data products in real-world production environments.						
Course Objectives						
1. To understand and design advanced architectures for handling and processing big data volumes, velocities, and varieties using distributed computing frameworks. 2. To master the principles and best practices of Machine Learning Operations (MLOps) for automated, reproducible, and continuously monitored AI lifecycles. 3. To comprehend and strategically leverage major cloud platform services for scalable AI/ML development, deployment, and management. 4. To architect and deploy complex, end-to-end AI-powered data products, ensuring scalability, reliability, and ethical governance in production environments.						
Course Outcomes						
CO1. Design and evaluate distributed data processing pipelines for high-throughput and real-time big data analytics. CO2. Develop and manage comprehensive MLOps strategies for automated model training, versioning, deployment, and monitoring. CO3. Propose and justify cloud-native architectures for AI/ML solutions, optimizing for performance, cost, and scalability. CO4. Lead the conceptual design and deployment of robust, production-grade AI-driven data products, considering all technical and ethical dimensions.						
Mapping						
	PO1	PO2	PO3	PO4	PSO1	PSO2
CO1	✓		✓		✓	✓
CO2	✓		✓		✓	✓
CO3	✓		✓	✓	✓	✓
CO4	✓	✓	✓	✓	✓	✓
Course Contents						
Unit 1	Big Data Architectures and Distributed Computing. Scalability, performance, fault tolerance, cost efficiency, data consistency, Conceptual architecture combining data lake flexibility with data warehouse structure and ACID properties, Real-time vs. Batch Processing Architectures (Lambda/Kappa architectures - conceptual differences), HDFS (detailed conceptual architecture, block management, NameNode/DataNode interactions, fault tolerance), Conceptual understanding of RDDs, directed acyclic graphs (DAGs), lazy evaluation, lineage, fault tolerance, Catalyst Optimizer (conceptual query optimization), integration with data catalogs (Hive Metastore - conceptual), Conceptual model for continuous processing, stateful					

	stream processing, exactly-once semantics, How ML algorithms are distributed and executed on Spark clusters, MongoDB (conceptual architecture for scalability), Cassandra (conceptual distributed architecture for high availability and write throughput), Kafka (conceptual architecture, topics, producers, consumers, broker roles for high-throughput streaming).
Unit 2	Machine Learning Operations (MLOps) in Practice. Data preparation, model development, experimentation, training, evaluation, model registration, deployment, monitoring, retraining, governance, Automation, Reproducibility, Continuous Experimentation, Continuous Delivery, Monitoring, Governance, Conceptual understanding of how to version datasets and manage reusable features (Feature Stores - conceptual role, benefits for consistency and reusability), Conceptual frameworks for tracking experiments (parameters, metrics, artifacts) and managing model versions in a central repository, Automated build, test, and deployment pipelines for ML models (conceptual integration with software CI/CD), Docker (conceptual understanding of building container images for consistent environments), Conceptual architecture for offline inference, API serving, microservices architecture, conceptual understanding of latency and throughput requirements, Deploying ML models on edge devices, A/B Testing and Canary Deployments for models (conceptual), Detecting data drift, concept drift, performance degradation (accuracy, bias, fairness metrics) in production, Strategies for triggered or scheduled retraining, continuous learning.
Unit 3	Cloud-Native AI and Data Analytics Architectures. IaaS, PaaS, SaaS models, elasticity, scalability, cost models, AWS (Amazon Web Services), Google Cloud Platform (GCP), Microsoft Azure, Object Storage (S3, GCS, Azure Blob Storage): Conceptual understanding of data lake implementation, tiered storage, data lifecycle management, Managed Data Warehouses (Redshift, BigQuery, Synapse Analytics): Conceptual architecture, MPP (Massively Parallel Processing) databases, Managed Stream Processing Services (Kinesis, Dataflow, Azure Stream Analytics): Conceptual role in real-time pipelines, AWS EMR, Google Cloud Dataproc, Azure HDInsight: Conceptual benefits for managed Spark/Hadoop clusters, Serverless Data Processing (AWS Glue, GCP Dataflow, Azure Data Factory/Synapse Pipelines - conceptual for ETL/ELT), Conceptual understanding of its various components (Notebook Instances, Training Jobs, Model Endpoints, Feature Store, Ground Truth, SageMaker Studio), Conceptual understanding of Notebooks, Training, Prediction, Explainable AI features, Vertex AI, Conceptual understanding of Workspaces, Experiment Tracking, Model Registry, Endpoints, MLOps integrations, Conceptual use of AWS Lambda, Google Cloud Functions, Azure Functions for event-driven ML inference and lightweight data processing, Kubernetes (EKS, GKE, AKS) for managing scalable ML microservices, Strategies for managing compute, storage, and service costs.
Unit 4	Designing and Deploying End-to-End AI Data Products. From ideation and problem framing to continuous improvement, Conceptual design patterns for data ingestion pipelines, model training workflows, inference services, and monitoring frameworks, Designing a large-scale fraud detection system: Data pipelines, real-time inference, model updates, Building a personalized recommendation engine: Data collection, offline training, online serving, feedback loops, Architecting a predictive maintenance system for industrial IoT: Sensor data ingestion, anomaly detection, forecasting, Developing an intelligent document processing system: NLP pipelines, image recognition, extraction, Roles and responsibilities of Data Engineers, ML Engineers, Data Scientists, DevOps, Product Managers, Conceptual understanding of

	APIs, microservices, and event-driven architectures for embedding AI into existing business processes.
Unit 5	Responsible AI in Production and Advanced Governance. How to build fairness, transparency, and accountability into MLOps pipelines, Continuous monitoring for bias drift, conceptual strategies for post-deployment debiasing, How to deliver model explanations to end-users and stakeholders, Conceptual understanding of Federated Learning and Differential Privacy in distributed systems, Data lineage, model lineage, auditability, compliance with regulations in a live system, Conceptual understanding of model security (adversarial attacks), data security in transit and at rest, access control, Their role in overseeing AI deployments.
Unit 6	Future Trends in AI for Data Science. Conceptual understanding of AutoML frameworks for model selection, hyperparameter tuning, and architecture search, Conceptual role in understanding true impact for better decision-making, Challenges and emerging patterns for deploying RL agents, Conceptual considerations for energy efficiency and environmental impact of large-scale AI models, Deploying AI models on resource-constrained devices, Conceptual shift towards decentralized data ownership and self-service data, Designing systems where human expertise complements AI predictions.
<i>This course follows an integrated lab model, wherein theoretical instruction is closely coupled with hands-on lab components. This approach ensures experiential learning, enabling students to apply AI concepts through practical exercises, coding assignments, and real-world datasets.</i>	
References	
1. McMahon A. <i>Machine Learning Engineering with Python: Manage the Production Life Cycle of Machine Learning Models Using MLOps with Practical Examples</i>. Packt Publishing, 1st Edition, 2021. 2. Singh P, Karuparti A. <i>Generative AI for Cloud Solutions: Architect Modern AI LLMs in Secure, Scalable, and Ethical Cloud Environments</i>. Packt Publishing, 1st Edition, 2023. 3. Osipov C. <i>MLOps Engineering at Scale: Deploying PyTorch Models on AWS</i>. Manning Publications, 1st Edition, 2020. 4. Walsh B. <i>Productionizing AI: How to Deliver AI B2B Solutions with Cloud and Python</i>. Apress, 1st Edition, 2020. 5. Lakshmanan V. <i>Data Science on the Google Cloud Platform: Implementing End-to-End Real-Time Data Pipelines: From Ingest to Machine Learning</i>. O'Reilly Media, 1st Edition, 2022.	

Specialization Bouquet 3: Generative AI

Course Code	AI-621					
Subject Name	Generative Artificial Intelligence Fundamentals					
Contact hours/Credit Scheme (L-T-P-C)	3-0-0-3					
Pre-requisites	None					
Course Objectives						
1. Develop a foundational understanding of generative AI techniques, including VAEs, GANs, and diffusion models.						
2. Apply mathematical concepts like probability, linear algebra, and information theory to AI model development.						
3. Implement generative models using Python libraries like TensorFlow and PyTorch.						
Course Outcomes						
CO1. Understand the core concepts of Generative AI and its evolution.						
CO2. Gain foundational knowledge of mathematical and neural network principles.						
CO3. Learn various generative models and their practical applications.						
CO4. Develop hands-on experience with Python-based implementations.						
CO5. Analyze ethical and societal challenges in AI						
Mapping						
	PO1	PO2	PO3	PO4	PSO1	PSO2
CO1		✓	✓		✓	✓
CO2	✓					
CO3		✓	✓			✓
CO4	✓			✓	✓	✓
CO5	✓		✓	✓	✓	
Course Contents						
Unit 1 (6 Hours)	Introduction to Generative AI: Overview of AI and Machine Learning: Understanding key AI concepts, types of machine learning, and introduction to generative AI. Evolution of Generative AI: From early rule-based systems to modern deep learning-based generative models. Key Applications and Use Cases: AI-driven content creation, synthetic media, data augmentation, and more.					
Unit 2 (8 Hours)	Probabilistic Foundations & Latent Variable Models: Probability distributions in generative modeling, Gaussian Mixture Models (GMMs), Variational inference, Hidden Markov Models (HMMs), Diffusion Models & Energy-based Models: Denoising Diffusion Probabilistic Models (DDPMs), Stable Diffusion and latent diffusion models, Energy-based models (EBMs).					
Unit 3 (7 Hours)	Generative Models: Overview: Autoencoders and Variational Autoencoders (VAEs): Theory, implementation, and applications. Generative Adversarial Networks (GANs): Architecture, training challenges, and practical					

	applications. Diffusion Models and Other Generative Approaches: Understanding diffusion-based models and their impact.
Unit 4 (6 Hours)	Transformers in Generative AI: Self-Attention Mechanism: Detailed breakdown of self-attention and its mathematical foundation. Transformer Architecture: Understanding key components like multi-head attention, positional encoding, and feedforward layers. BERT vs GPT: Differences in objectives, architectures, and applications.
Unit 5 (9 Hours)	Ethics and Challenges in Generative AI: Bias, Fairness, and Explainability: Ethical concerns, biases in AI models, and explainability techniques. Societal Implications: AI-generated deepfakes, misinformation, and their impact. Mitigation Strategies: Approaches to ensure responsible AI use.
Hands-on Assignments & Integrated Lab: Implementing VAEs and GANs using Python: Writing and training models from scratch. Training Simple Generative Models: Working with TensorFlow and PyTorch implementations.	
References	
1. Foster D. <i>Generative Deep Learning</i>. O'Reilly Media 2008 2. Babcock J, Bali R. <i>Generative AI with Python and TensorFlow</i>. Packt Publishing, 2021 3. Goodfellow I, Bengio Y, Courville A. <i>Deep Learning</i>. MIT Press. 2008 4. Bishop C. <i>Pattern Recognition and Machine Learning</i>. Springer New York.2011 5. Burkov A. <i>The Hundred-Page Machine Learning Book</i>. O'Reilly Media. 2006 6. Tunstall L, Werra LV, Wolf T. <i>Natural Language Processing with Transformers</i>. O'Reilly Media. 2009	

Course Code	AI-622					
Subject Name	Large Language Models					
Contact hours/Credit Scheme (L-T-P-C)	3-0-0-3					
Pre-requisites	Basics of Machine Learning, Deep Learning, and Python Programming					
Course Objectives						
1. To introduce foundational principles behind Large Language Models (LLMs) like GPT, PaLM, LLaMA, Claude, and DeepSeek. 2. To explore prompt engineering, fine-tuning, and in-context learning for performance enhancement. 3. To study architecture deployment and memory solutions for LLMs and diffusion models. 4. To enable students to build applications using LLMs, vector databases, and evaluate outputs using suitable metrics.						
Course Outcomes						
CO1. Identify and explain various state-of-the-art LLMs and their architectures including GPT, PaLM, LLaMA, DeepSeek, and Claude.						
CO2. Implement prompt engineering, in-context learning, and fine-tuning strategies in LLM-driven applications.						
CO3. Deploy and serve LLMs and manage memory-efficient storage using vector databases like FAISS, Pinecone, and Weaviate.						
CO4. Evaluate LLM-based applications such as chatbots, code generation, and summarization using modern evaluation metrics.						
Mapping						
	PO1	PO2	PO3	PO4	PSO1	PSO2
CO1	✓				✓	✓
CO2		✓	✓	✓	✓	✓
CO3	✓	✓	✓	✓	✓	✓
CO4	✓	✓	✓	✓	✓	✓
Course Contents						
Unit 1 (10 Hrs)	Foundations of LLMs Introduction to LLMs: Evolution and Need, Transformer Architecture Refresher, Overview of prominent LLMs: GPT, PaLM, LLaMA, Claude, DeepSeek, Capabilities and Limitations of LLMs, Use cases and real-world relevance					
Unit 2 (12 Hrs)	Prompting and Learning Strategies Prompt Engineering Techniques, Fine-tuning vs In-Context Learning, Chain-of-Thought Prompting and Few-shot Learning, Evaluation of LLM outputs: BLEU, ROUGE, BERTScore, Best practices in model tuning and responsible use					
Unit 3 (9 Hrs)	Serving and Infrastructure Retrieval-Augmented Generation (RAG), Serving LLMs and Diffusion Models at Scale, Introduction to Vector Databases: FAISS, Pinecone, Weaviate, Memory and latency optimization strategies, Integration with external APIs and tools					
Unit 4 (9 Hrs)	Applications of LLMs Building Conversational Agents and Chatbots, Code Generation (e.g., GitHub Copilot), Document Summarization and Question-Answering Systems, Language-based Automation Workflows, Ethical considerations and hallucination handling					

References

1. Magnus Ekman. *Learning Deep Learning: Theory and Practice of Neural Networks, Computer Vision, Natural Language Processing, and Transformers using TensorFlow*, Addison-Wesley 2021.
2. Anirudh Koul, Siddha Ganju. *Practical Generative AI: Tools and Applications Using ChatGPT and OpenAI APIs*, O'Reilly Media, 2023
3. OpenAI Cookbook and Documentation (<https://platform.openai.com/docs>)
4. Google Research Papers on PaLM, BERT, and T5
5. Hugging Face Transformers Library Documentation
6. Pinecone, FAISS, Weaviate Official Docs and Tutorials

e-sources:

7. <https://elearn.nptel.ac.in/shop/iit-workshops/completed/leveraging-generative-ai-for-teaching-programming-course/s/?v=c86ee0d9d7ed>
8. <https://elearn.nptel.ac.in/shop/iit-workshops/completed/introduction-to-language-models/?v=c86ee0d9d7ed>

Course Code	AI-623					
Subject Name	Applied Generative AI and Prompt Engineering					
Contact hours/Credit Scheme (L-T-P-C)	3-0-0-3					
Pre-requisites	None					
Course Objectives						
1. Understand and explore real-world applications of Generative AI across various domains including art, education, healthcare, and entertainment.						
2. Gain practical skills in prompt engineering for text, image, and multimodal generative models using industry-standard APIs and tools.						
3. Design, implement, and optimize generative AI pipelines by integrating pre-trained models and third-party platforms						
Course Outcomes						
CO1. Demonstrate the ability to identify and scope real-world problems that can be solved using Generative AI technologies.						
CO2. Create and refine effective prompts for different generative tasks (text generation, text-to-image, image-to-text, etc.) using tools like GPT-4 and DALL·E						
CO3. Develop and deploy simple generative AI applications by integrating APIs and pre-trained models.						
CO4. Evaluate generative outputs using objective and subjective metrics, and iteratively optimize prompt design or model parameters						
CO5. Collaboratively design and present a complete applied GenAI project with documented use cases, implementation workflow, and deployment strategies.						
Mapping						
	PO1	PO2	PO3	PO4	PSO1	PSO2
CO1		✓	✓		✓	✓
CO2	✓			✓		
CO3	✓	✓	✓	✓		✓
CO4				✓	✓	✓
CO5		✓	✓	✓	✓	
Course Contents						
Unit 1 (6 Hours)	Real-world Applications of Generative AI, AI in art, gaming, and education. Industry-specific use cases: Finance, healthcare, and more.					
Unit 2 (8 Hours)	Prompt Engineering. Designing effective prompts for text-to-image and text generation. Iterative prompt refinement and evaluation.					
Unit 3 (7 Hours)	Frameworks and APIs, Using Hugging Face, OpenAI, and Stability AI APIs for rapid prototyping.					
Unit 4 (6 Hours)	Transformer-based Models, GPT and language generation. Multimodal transformers (e.g., DALL-E, Stable Diffusion).					
Unit 5 (9 Hours)	Optimization Techniques. Transfer learning and fine-tuning pre-trained generative models. Reinforcement learning in Generative AI.					

Hands-on Assignments & Integrated Lab: Developing prompts for DALL-E and GPT-4 to achieve desired outputs. Creating a text-to-image generation pipeline using pre-trained models. Capstone Project: Identifying a problem statement and initiating a project using generative AI

References

1. Sansevier O, Cuenca P, Passos A, Whitaker J. *Hands-On Generative AI with Transformers and Diffusion Models*. O'Reilly Media, 2012
2. Kharwal A. *From ML Algorithms to GenAI & LLMs: Master ML Algorithms and Generative AI & LLMs with Python from scratch*. Bluerose Publishers Pvt. Ltd. 2024.
3. Jha SK. *GenAI Foundations: LLMS From Fundamental to Future Frontiers*. Notion Press 2025.
4. Sabesan K, Sivagamisundari, Dutta N. *Generative AI for Everyone: Deep learning, NLP, and LLMs for creative and practical applications*. Bpb Publications 2025.
5. Ozdimer S. *Quick Start Guide to Large Language Models*. Addison-Wesley Professional 2024.
6. Pai S. *Designing Large Language Model Applications: A Holistic Approach to LLMs*. O'Reilly Media 2025.

Course Code	AI-624					
Subject Name	Agentic AI					
Contact hours/Credit Scheme (L-T-P-C)	3-0-0-3					
Pre-requisites	Basics of Machine Learning, Deep Learning, and Python Programming					
Course Objectives						
1. Introduce students to foundational agent architectures and their decision-making mechanisms.						
2. Explore reinforcement learning, meta-learning, and agent learning paradigms.						
3. Study interactive agent capabilities like tool usage, perception, and external API interaction.						
4. Enable development of multi-agent systems and language-based instruction-following agents.						
Course Outcomes						
CO1. Explain and analyze various agent architectures including reactive, deliberative, and hybrid.						
CO2. Apply reinforcement learning and meta-learning methods in agent training.						
CO3. Develop agents capable of using tools and following natural language instructions.						
CO4. Design multi-agent systems capable of collaboration and game-theoretic decision-making.						
Mapping						
	PO1	PO2	PO3	PO4	PSO1	PSO2
CO1		✓	✓			✓
CO2	✓	✓	✓	✓	✓	✓
CO3	✓	✓	✓		✓	✓
CO4	✓	✓	✓	✓	✓	✓
Course Contents						
Unit 1 (8 HRS)	Introduction to Agentic AI and Intelligent Agents, Agent Architectures: Reactive, Deliberative, and Hybrid Models, Belief-Desire-Intention (BDI) Framework, Case Studies in Classic AI Agents					
Unit 2 (15 HRS)	Reinforcement Learning: Q-Learning, Policy Gradients, Deep Q-Networks (DQN), Meta-Learning for Agent Adaptation, Tool-using Agents: Concept and Use-cases (AutoGPT, LangChain), External API Interaction and Web-enabled Agents, Agent Simulators and Open-Source Platforms (e.g., OpenAI Gym, PettingZoo)					
Unit 3 (9 HRS)	Natural Language Interfaces: Prompting and Instruction Following, Language-based Planning and Chain-of-Thought Reasoning, Embodied Agents: Perception-Action Loops and Environmental Interaction, Examples from Robotics and Virtual Assistants					
Unit 4 (8 HRS)	Multi-Agent Systems: Architectures and Coordination, Game-Theoretic Decision Making: Nash Equilibrium, Cooperative Games, Communication					

	Among Agents, Ethical and Social Implications of Agentic AI
References	
1. David Foster. <i>Generative Deep Learning: Teaching Machines to Paint, Write, Compose, and Play</i>, O'Reilly Media 2023 2. Magnus Ekman. <i>Learning Deep Learning: Theory and Practice of Neural Networks, Computer Vision, Natural Language Processing, and Transformers using TensorFlow</i>, Addison-Wesley 2021). 3. Josh Kalin. <i>Generative Adversarial Networks Cookbook: Over 100 recipes to build generative models using Python, TensorFlow, and Keras</i>. Packt Publishing. 2021. 4. Anirudh Koul, Siddha Ganju. <i>Practical Generative AI: Tools and Applications Using ChatGPT and OpenAI APIs</i>, O'Reilly Media, 2023 e-sources: 5. https://elearn.nptel.ac.in/shop/iit-workshops/completed/leveraging-generative-ai-for-teaching-programming-course s/?v=c86ee0d9d7ed 6. https://elearn.nptel.ac.in/shop/iit-workshops/completed/introduction-to-language-models/?v=c86ee0d9d7ed	

Specialization Bouquet 4: AI for Signal and Image Processing

Course Code	AI-631					
Subject Name	Fundamentals of Signal and Image Processing for AI					
Contact hours/Credit Scheme (L-T-P-C)	3-0-0-3					
Pre-requisites	None					
Course Objectives						
1. Understand the foundational concepts of 1D signal processing and 2D image processing.						
2. Learn mathematical tools for analyzing signals and images in both time and frequency domains.						
3. Study techniques for filtering, transforming, compressing, and enhancing signals and images.						
4. Prepare students for advanced AI applications in signal and image domains.						
Course Outcomes						
CO1. Explain the properties of signals and images and perform basic transformations.						
CO2. Apply techniques such as DFT, FFT, and filtering to signals and images.						
CO3. Analyze images using spatial and frequency domain techniques.						
CO4. Evaluate the effectiveness of compression and enhancement techniques.						
Mapping						
	PO1	PO2	PO3	PO4	PSO1	PSO2
CO1	✓		✓		✓	✓
CO2		✓	✓	✓		✓
CO3	✓	✓	✓		✓	
CO4	✓		✓	✓		✓
Course Contents						
Unit 1	Introduction to Signals and Systems, Continuous and discrete signals, Sampling and aliasing, Linear time-invariant (LTI) systems, Convolution, and correlation.					
Unit 2	Frequency Domain Analysis, Discrete Fourier Transform (DFT), Fast Fourier Transform (FFT), Time-frequency representations (STFT), Wavelet transforms (Haar, Daubechies)					
Unit 3	Image Representation and Enhancement, Digital image model, intensity transformations, Histogram equalization and matching, Spatial filtering: smoothing and sharpening, Edge detection: Sobel, Prewitt, Canny.					
Unit 4	Image and Signal Filtering, FIR and IIR filters, High-pass and low-pass filters, Morphological operations (dilation, erosion, opening, closing), Median, Gaussian, and bilateral filters.					
Unit 5	Compression and Feature Extraction, Run-length encoding, JPEG and PNG basics, PCA (Principal Component Analysis), HOG (Histogram of Oriented Gradients), SIFT/SURF (conceptual overview).					
Hands-On Learning and Integrated lab: Applications and Mini Project Audio signal enhancement Image quality assessment, ECG denoising.						
References						
1. Oppenheim AV, Vergheese GC. <i>Signals, Systems and Inference</i> . Pearson Education , 2018.						
2. Proakis JG. <i>Digital Signal Processing: Principles, Algorithms and Applications</i> . Prentice Hall , 2007						
3. Gonzalez RC, Woods RE. <i>Digital Image Processing</i> . Pearson . 2018						
4. Downey AB. <i>Think DSP: Digital Signal Processing in Python</i> . O'Reilly Media , 2016.						
5. Research Articles from the related popular journals						

Course Code	AI-632					
Subject Name	Deep Learning for Signal and Image Processing					
Contact hours/Credit Scheme (L-T-P-C)	3-0-0-3					
Pre-requisites	None					
Course Objectives						
1. Understand deep learning architectures for analyzing time series and images.						
2. Explore CNNs, RNNs, LSTMs, attention models, and transformers for multimodal data.						
3. Apply AI to advanced tasks like segmentation, synthesis, detection, and classification in biomedical and visual domains.						
4. Deploy deep learning models in signal and image applications with attention to performance and explainability.						
Course Outcomes						
CO1. Design and implement deep learning pipelines for complex signal and image tasks.						
CO2. Apply and fine-tune architectures like CNNs, RNNs, and Transformers.						
CO3. Use pre-trained models and fine-tune on domain-specific datasets.						
CO4. Analyze and interpret model predictions using attention maps and saliency techniques.						
Mapping						
	PO1	PO2	PO3	PO4	PSO1	PSO2
CO1	✓		✓	✓	✓	✓
CO2		✓			✓	✓
CO3	✓				✓	✓
CO4	✓	✓	✓	✓		✓
Course Contents						
Unit 1 (6 Hours)	Deep Learning Foundations for Signals and Images, CNNs for image recognition, RNNs, LSTMs, GRUs for time series, Data pipelines and augmentation techniques.					
Unit 2 (5 hours)	Domain-specific Deep Models, Signal: EEG seizure detection, ECG classification, Image: Medical imaging (X-ray, MRI), satellite image classification, Custom model architectures and hyperparameter tuning.					
Unit 3 (7 Hours)	Advanced Architectures, Transformers for signal and image modeling, Attention mechanisms and temporal models, Autoencoders and Variational Autoencoders (VAEs).					
Unit 4 (4 Hours)	Generative Models and Segmentation, Generative Adversarial Networks (GANs) for synthetic data, U-Net for image segmentation, Real-time applications: speech synthesis, denoising autoencoders.					
Unit 5 (7 Hours)	Model Deployment and Explainability, Saliency maps, Grad-CAM, attention visualization, Model compression and optimization, Deployment tools: TensorFlow Lite, ONNX, Flask/FastAPI					
Unit 6 (6 Hours)	Human Motion Tracking: Image feature representation, Dimension reduction and Movement dynamics learning. Human action recognition: Discriminative Gaussian Process dynamic model. Human Interaction recognition: Learning human activity, Track-body Synergy framework. Multi-camera calibration and global trajectory fusion: Non-overlapping and overlapping cameras					
Unit 7 (3 Hours)	Behavioral Analysis and Activity Recognition: Event Modeling- Behavioral Analysis- Human Activity Recognition-Complex Activity Recognition- Activity modeling using 3D shape, Video summarization, Shape based activity models, Suspicious Activity Detection.					

Hands-On Learning and Integrated lab: Python, PyTorch, TensorFlow, Keras, OpenCV, scikit-image, Albumentations, Datasets: ISIC, DRISHTI-GS, Pascal VOC, COCO, EuroSAT.

References

1. Elgendy M. *Deep Learning for Vision Systems*. [Manning](#), 2020
2. Solem JE. *Programming Computer Vision with Python: Tools and Algorithms for Analyzing Images*. [O'Reilly Media](#), 2012
3. Camastra F, Vinciarelli A. *Machine Learning for Audio, Image and Video Analysis*, Springer Nature. 2015.
4. Ma Y, Qian G. *Intelligent Video Surveillance: Systems and Technology*, CRC Press, 2009.
5. Nilsson F, Axis C. *Intelligent Network Video: Understanding Modern Video Surveillance Systems*, CRC Press. 2017.
6. Research Articles from the related popular journals

Course Code	AI-633					
Subject Name	Machine Learning for Signal and Image Processing					
Contact hours/Credit Scheme (L-T-P-C)	3-0-0-3					
Pre-requisites	None					
Course Objectives						
1. Understand the foundations of signal and image representations for ML.						
2. Learn classical machine learning algorithms for classification, regression, and clustering on signals and images.						
3. Perform feature engineering and extraction using domain knowledge in signal and image processing.						
4. Apply ML to applications in speech, biomedical signals (e.g., ECG, EEG), and 2D image recognition.						
Course Outcomes						
CO1. Extract and preprocess features from raw signal and image datasets.						
CO2. Train and validate machine learning models for classification and regression tasks on multimodal data.						
CO3. Apply dimensionality reduction and model interpretability tools.						
CO4. Design end-to-end ML pipelines for real-world signal/image problems.						
Mapping						
	PO1	PO2	PO3	PO4	PSO1	PSO2
CO1	✓		✓		✓	✓
CO2		✓	✓	✓		✓
CO3	✓	✓	✓		✓	
CO4	✓		✓	✓		✓
Course Contents						
Unit 1 (8 hours)	Basics of Signal and Image Data. Signal modalities: 1D physiological signals (ECG, EEG, EMG), speech, Image modalities: grayscale, RGB, medical images Sampling, quantization, normalization, segmentation.					
Unit 2 (9 hours)	Feature Engineering and Preprocessing. Time-domain and frequency-domain features for signals. Histogram of oriented gradients (HOG), color histograms for images. PCA, t-SNE, data augmentation and denoising.					
Unit 3 (8 hours)	Machine Learning Models, Supervised Learning: SVM, Random Forest, kNN, Logistic Regression, Unsupervised Learning: Clustering, dimensionality reduction, Time-series ML with traditional models (ARIMA, HMMs).					
Unit 4 (9 Hours)	Applications in Signal and Image Domains, Speech emotion recognition, Biomedical signal classification (ECG arrhythmia, EEG seizure), Image recognition and texture classification.					
Unit 5 (6 Hours)	Evaluation and Explainability, Cross-validation, confusion matrices, ROC curves, SHAP, LIME for classical model explainability, Feature importance analysis.					
Hands-On Learning and Integrated lab: Applications and Mini Project Audio signal enhancement Image quality assessment, ECG denoising.						
References						
1. Brownlee J. <i>Introduction to Time Series Forecasting with Python</i> . Jason Brownlee . 2017.						
2. Murohy KP. <i>Machine Learning: A Probabilistic Perspective</i> . MIT Press , 2012						
3. Rangayyan RM. <i>Biomedical Signal Analysis</i> . Wiley . 2015						
4. Downey AB. <i>Think DSP: Digital Signal Processing in Python</i> . O'Reilly Media , 2016.						
5. Research Articles from the related popular journals						
6. Scikit-learn and OpenCV documentation						

Course Code	AI-634					
Subject Name	Advanced AI Applications in Signal and Image Domain					
Contact hours/Credit Scheme (L-T-P-C)	3-0-0-3					
Pre-requisites	None					
Course Objectives						
1. Understand the diverse application areas of AI in time series and imaging domains, including healthcare, finance, energy, and satellite data. 2. Gain knowledge of clinical and technical foundations for AI in 2D and 3D medical imaging and understand regulatory deployment aspects. 3. Learn explainable AI techniques tailored for time series and sequence models, with tools like SHAP and attention mechanisms. 4. Explore real-time deployment of AI models using serving frameworks and streaming infrastructures.						
Course Outcomes						
CO1. Apply AI models to real-world time series problems such as seizure detection, brain-computer interfacing, and speech emotion classification.						
CO2. Design and implement AI pipelines for 2D and 3D medical image analysis with clinical applicability.						
CO3. Integrate explainable AI methods into time series modeling for transparency and trust.						
CO4. Implement video analytics tasks including segmentation, object tracking, and key-frame extraction using modern AI algorithms.						
Mapping						
	PO1	PO2	PO3	PO4	PSO1	PSO2
CO1	✓			✓	✓	✓
CO2		✓			✓	
CO3		✓				✓
CO4	✓			✓		✓
Course Contents						
Unit 1 (10 Hours)	Applications of AI in key Time Series analysis domains. Finance (stock prediction, volatility modeling), Healthcare (Electrocardiogram - ECG, Electroencephalogram - EEG pattern analysis). Internet of Things (IoT) and sensor analytics, Energy (load forecasting). Case study: Seizure detection using EEG. Brain computer Interfacing. Speech emotion classification. Domain-specific AI project on real-time signal data.					
Unit 2 (8 Hours)	AI for 2D Medical Imaging: Introduction to AI for 2D medical imaging, Clinical foundations of AI for 2D medical imaging, 2D medical imaging exploratory data analysis, Classification models of 2D medical images, Translating AI algorithms for clinical settings with the FDA. AI for 3D Medical Imaging: Introduction to AI for 3D medical imaging, 3D medical imaging - Clinical fundamentals, 3D medical imaging exploratory data analysis, 3D medical images deep learning methods, Deploying AI algorithms in the real worlds.					
Unit 3 (5 Hours)	Explainability in AI for Time Series Feature attribution using SHapley Additive exPlanations (SHAP). Attention visualization in sequence models. Deployment of Time Series Models, Model serving: TensorFlow Serving, FastAPI. Real-time inference with streaming frameworks: Apache Kafka, Apache Spark Streaming. Model monitoring and retraining pipelines.					
Unit 4	Applications of AI in key imaging domains. Medical imaging (X-ray, MRI, CT scan,					

(9 Hours)	Ultrasound Image classification). Satellite image segmentation. Real-time object tracking. Drone Image processing and surveillance.
Unit 5 (8 Hours)	Video Segmentation and Key Frame Extraction: Introduction, Applications of Video Segmentation, Shot Boundary Detection, Pixel-based Approaches, Block-based Approaches, Histogram-based Approaches, Clustering-based Approaches, Performance Measures, Shot Boundary Detection, Key-frame Extraction.
Hands-On Learning and Integrated lab: Python, PyTorch, TensorFlow, Keras, OpenCV, scikit-image, Albumentations, Datasets: ISIC, DRISHTI-GS, Pascal VOC, COCO, EuroSAT.	
References	
1. Brownlee J. <i>Introduction to Time Series Forecasting with Python</i>. Jason Brownlee. 2017 2. Monlar C. <i>Interpretable Machine Learning: A Guide for Making Black Box Models Explainable</i>. Leanpub. 2020. 3. Shen D, Greenspan H, Zhou SK. <i>Deep Learning for Medical Image Analysis</i>. Elsevier Science, 2017 4. Louk A, Ganju S, Kasam M. <i>Practical Deep Learning for Cloud, Mobile, and Edge: Real-World AI & Computer-Vision Projects Using Python, Keras & TensorFlow</i>. O'Reilly Media. 2019. 5. Research Articles from the related popular journals	

Proposal for New Academic Course

Name of the Centre: Centre for Artificial Intelligence

1	Title of the Course	Machine Learning for Time Series Forecasting											
2	Credit Structure	<i>Lectures (L)</i>	<i>Tutorials (T)</i>	<i>Practicals (P)</i>	<i>Total Credits (C)</i>								
		3	0	0	3								
3	Academic Programme for which course is applicable	☐ Undergraduate ☒ Postgraduate ☐ Ph.D.											
4	Type of Course	☒ Theory ☐ Seminar ☐ Lab											
5	Course Tag	☒ Core ☐ Department Elective ☐ General Elective ☐ Minor											
6	Justification for Introduction of course	This course bridges classical and AI-driven approaches to time series analysis, covering forecasting, anomaly detection, and sequence modeling using ML and deep learning tools. Students gain hands-on experience with real-world datasets from domains such as healthcare, finance, and climate. With increasing demand for predictive intelligence across industries, this subject equips students with in-demand, application-oriented skills and prepares them for roles in AI engineering, data science, and applied research.											
7	Contents of the course & Tentative Lecture Plan	Attached within Main Curriculum											
8	Texts and References (Minimum 5 – Maximum 8) (Complete name of Author/ Title/ Edition/ Publisher/ Volume, Web references/ e-references, year of publication, etc.)	1. Brockwell PJ, Davis RA. <i>Introduction to Time Series and Forecasting</i>. Springer International Publishing, 2013. 2. Shmueli G. <i>Practical Time Series Forecasting: A Hands-On Guide</i>. Axelrod Schnall Publishers, 2016. 3. <u>Brownlee J</u>. <i>Introduction to Time Series Forecasting with Python</i>. <u>Jason Brownlee</u>. 2017 4. <u>Lazzeri F</u>. <i>Machine Learning for Time Series Forecasting with Python</i>. <u>Wiley</u>. 2020 5. Chaudhuri TD. <i>Analysis and Forecasting of Financial Time Series Using R: Models and Applications</i>. Scholars' Press, 2017.											
9	Names of Instructors (Require the names of at least two permanent faculty members of NITJ for core course and one permanent faculty member of NITJ for elective course.)	<table border="1" style="width: 100%; border-collapse: collapse;"> <tr> <th colspan="2" style="text-align: left;">Name of Instructor & Department</th> </tr> <tr> <td style="width: 5%; text-align: center;">1.</td><td>Dr Arun Khosla, Professor, Electronics and Communication Engineering</td> </tr> <tr> <td style="text-align: center;">2.</td><td>Dr OP Verma, Assistant Professor, Instrumentation and Control Engineering.</td> </tr> <tr> <td style="text-align: center;">3.</td><td>Dr KC Sharma, Assistant Professor, Electrical Engineering.</td> </tr> </table>				Name of Instructor & Department		1.	Dr Arun Khosla, Professor, Electronics and Communication Engineering	2.	Dr OP Verma, Assistant Professor, Instrumentation and Control Engineering.	3.	Dr KC Sharma, Assistant Professor, Electrical Engineering.
Name of Instructor & Department													
1.	Dr Arun Khosla, Professor, Electronics and Communication Engineering												
2.	Dr OP Verma, Assistant Professor, Instrumentation and Control Engineering.												
3.	Dr KC Sharma, Assistant Professor, Electrical Engineering.												
10	Existing courses with overlapping content along with %age overlapping (Offered by the same or other departments)	5% overlap with CS-503 Advanced Databases and Data Mining MTech in Information Security, Computer Science and Engineering											
11	Other departments to whom the course may be relevant (if any):												

The above proposal for a new course is found to be acceptable by (DUGC/ DPGC) in its meeting held on _____. The committee recommends this course proposal for consideration of SUGC / SPGC.

**Signature of the Chairman,
DUGC/ DPGC**

Proposal for New Academic Course

Name of the Centre: Centre for Artificial Intelligence

1	Title of the Course	Machine Learning Laboratory											
2	Credit Structure	<i>Lectures (L)</i>	<i>Tutorials (T)</i>	<i>Practicals (P)</i>	<i>Total Credits (C)</i>								
		0	0	3	2								
3	Academic Programme for which course is applicable	☐ Undergraduate ☒ Postgraduate ☐ Ph.D.											
4	Type of Course	☐ Theory ☐ Seminar ☒ Lab											
5	Course Tag	☒ Core ☐ Department Elective ☐ General Elective ☐ Minor											
6	Justification for Introduction of course	The <i>Machine Learning Laboratory</i> equips students with essential hands-on skills to apply ML algorithms on real-world data. It complements theoretical learning by enabling practical implementation, tuning, and evaluation of models using tools like Scikit-learn and TensorFlow. Through this lab, students gain experience in data preprocessing, model training, and performance analysis while exploring key concepts such as overfitting, bias-variance tradeoff, and ML pipelines. This practical exposure prepares students for advanced AI coursework and careers in research or industry, where applied ML skills are in high demand.											
7	Contents of the course & Tentative Lecture Plan	Attached within Main Curriculum											
8	Texts and References (Minimum 5 – Maximum 8) <i>(Complete name of Author/ Title/ Edition/ Publisher/ Volume, Web references/ e-references, year of publication, etc.)</i>	1. Nilsson NJ. <i>Quest for Artificial Intelligence</i>. Cambridge University Press, 2010. 2. Mitchell TM. <i>Machine Learning</i>. Mc Graw Hill Education, 1997. 3. Alpaydin E, <i>Introduction to Machine Learning</i>. MIT Press, 2014. 4. Flaszinski M. <i>Introduction to Artificial Intelligence</i>. SPRINGER 2017. 5. Clark RM. <i>Intelligence analysis</i>. Sage Publications, 2012.											
9	Names of Instructors <i>(Require the names of at least two permanent faculty members of NITJ for core course and one permanent faculty member of NITJ for elective course.</i>	<table border="1" style="width: 100%; border-collapse: collapse;"> <tr> <td colspan="2" style="text-align: center;">Name of Instructor & Department</td> </tr> <tr> <td style="width: 5%; text-align: center;">1.</td> <td>Dr Prashant Kumar, Assistant Professor, Computer Science and Engineering</td> </tr> <tr> <td style="text-align: center;">2.</td> <td>Dr OP Verma, Assistant Professor, Instrumentation and Control Engineering.</td> </tr> <tr> <td style="text-align: center;">3.</td> <td>Dr Urvashi, Assistant Professor, Computer Science and Engineering</td> </tr> </table>				Name of Instructor & Department		1.	Dr Prashant Kumar, Assistant Professor, Computer Science and Engineering	2.	Dr OP Verma, Assistant Professor, Instrumentation and Control Engineering.	3.	Dr Urvashi, Assistant Professor, Computer Science and Engineering
Name of Instructor & Department													
1.	Dr Prashant Kumar, Assistant Professor, Computer Science and Engineering												
2.	Dr OP Verma, Assistant Professor, Instrumentation and Control Engineering.												
3.	Dr Urvashi, Assistant Professor, Computer Science and Engineering												
10	Existing courses with overlapping content along with %age overlapping <i>(Offered by the same or other departments)</i>	20% overlap with CS-625 Advanced Machine Learning Laboratory MTech in Data Science & Engineering, Computer Science and Engineering											
11	Other departments to whom the course may be relevant (if any):												

The above proposal for a new course is found to be acceptable by (DUGC/ DPGC) in its meeting held on _____. The committee recommends this course proposal for consideration of SUGC / SPGC.

**Signature of the Chairman,
DUGC/ DPGC**

Proposal for New Academic Course

Name of the Centre: Centre for Artificial Intelligence

1	Title of the Course	Concurrent Programming Laboratory											
2	Credit Structure	<i>Lectures (L)</i>	<i>Tutorials (T)</i>	<i>Practicals (P)</i>	<i>Total Credits (C)</i>								
		3	0	0	3								
3	Academic Programme for which course is applicable	☐ Undergraduate ☒ Postgraduate ☐ Ph.D.											
4	Type of Course	☒ Theory ☐ Seminar ☐ Lab											
5	Course Tag	☒ Core ☐ Department Elective ☐ General Elective ☐ Minor											
6	Justification for Introduction of course	The <i>Concurrent Programming</i> course introduces the principles and models required to design systems that perform multiple computations simultaneously. As modern processors and distributed systems rely heavily on concurrency for efficiency, this course equips students with foundational knowledge of threads, synchronization, deadlocks, and concurrent data structures. It prepares them to develop robust, scalable software for multi-core, parallel, and distributed environments, which is crucial in domains like operating systems, cloud computing, and high-performance applications.											
7	Contents of the course & Tentative Lecture Plan	Attached within Main Curriculum											
8	Texts and References (Minimum 5 – Maximum 8) (Complete name of Author/ Title/ Edition/ Publisher/ Volume, Web references/ e-references, year of publication, etc.)	1. Sanders J, Kandrot E. <i>CUDA by Example</i>. Pearson Education, 2010. 2. Kurk DB, Hwu WW. <i>Programming Massively Parallel Processors, A Hands-on Approach</i>. Morgan Kaufmann, 2010 3. Shane C. <i>CUDA Programming</i>. Elsevier Science. 2012 4. NVIDIA CUDA Toolkit Documentation 5. CUDA samples and SDKs from NVIDIA											
9	Names of Instructors (Require the names of at least two permanent faculty members of NITJ for core course and one permanent faculty member of NITJ for elective course.)	<table border="1" style="width: 100%; border-collapse: collapse;"> <tr> <th colspan="2" style="text-align: left;">Name of Instructor & Department</th> </tr> <tr> <td style="width: 5%; text-align: center;">1.</td> <td>Dr Prashant Kumar, Assistant Professor, Computer Science and Engineering</td> </tr> <tr> <td style="text-align: center;">2.</td> <td>Dr OP Verma, Assistant Professor, Instrumentation and Control Engineering.</td> </tr> <tr> <td style="text-align: center;">3.</td> <td></td> </tr> </table>				Name of Instructor & Department		1.	Dr Prashant Kumar, Assistant Professor, Computer Science and Engineering	2.	Dr OP Verma, Assistant Professor, Instrumentation and Control Engineering.	3.	
Name of Instructor & Department													
1.	Dr Prashant Kumar, Assistant Professor, Computer Science and Engineering												
2.	Dr OP Verma, Assistant Professor, Instrumentation and Control Engineering.												
3.													
10	Existing courses with overlapping content along with %age overlapping (Offered by the same or other departments)	No Overlap.											
11	Other departments to whom the course may be relevant (if any):												

The above proposal for a new course is found to be acceptable by (DUGC/ DPGC) in its meeting held on _____. The committee recommends this course proposal for consideration of SUGC / SPGC.

**Signature of the Chairman,
DUGC/ DPGC**

Proposal for New Academic Course

Name of the Centre: Centre for Artificial Intelligence

1	Title of the Course	Reinforcement Learning											
2	Credit Structure	<i>Lectures (L)</i>	<i>Tutorials (T)</i>	<i>Practicals (P)</i>	<i>Total Credits (C)</i>								
		0	0	3	2								
3	Academic Programme for which course is applicable	☐ Undergraduate ☒ Postgraduate ☐ Ph.D.											
4	Type of Course	☒ Theory ☐ Seminar ☐ Lab											
5	Course Tag	☒ Core ☐ Department Elective ☐ General Elective ☐ Minor											
6	Justification for Introduction of course	The <i>Reinforcement Learning</i> course introduces students to the foundational concepts of learning through interaction and delayed rewards, which are central to decision-making in uncertain environments. It covers key topics such as Markov Decision Processes, value functions, policy optimization, and Q-learning. As RL drives advancements in robotics, game AI, recommendation systems, and autonomous systems, this course equips students with the theoretical tools to model and solve complex sequential decision problems, preparing them for cutting-edge research and real-world AI applications.											
7	Contents of the course & Tentative Lecture Plan	Attached within Main Curriculum											
8	Texts and References (Minimum 5 – Maximum 8) <i>(Complete name of Author/ Title/ Edition/ Publisher/ Volume, Web references/ e-references, year of publication, etc.)</i>	1. Sanders J, Kandrot E. <i>CUDA by Example</i>. Pearson Education, 2010. 2. Kurk DB, Hwu WW. <i>Programming Massively Parallel Processors, A Hands-on Approach</i>. Morgan Kaufmann, 2010 3. Shane C. <i>CUDA Programming</i>. Elsevier Science. 2012 4. NVIDIA CUDA Toolkit Documentation 5. CUDA samples and SDKs from NVIDIA											
9	Names of Instructors <i>(Require the names of at least two permanent faculty members of NITJ for core course and one permanent faculty member of NITJ for elective course.</i>	<table border="1" style="width: 100%; border-collapse: collapse;"> <tr> <th colspan="2" style="text-align: left;">Name of Instructor & Department</th> </tr> <tr> <td style="width: 5%; text-align: center;">1.</td> <td>Dr Prashant Kumar, Assistant Professor, Computer Science and Engineering</td> </tr> <tr> <td style="text-align: center;">2.</td> <td>Dr OP Verma, Assistant Professor, Instrumentation and Control Engineering.</td> </tr> <tr> <td style="text-align: center;">3.</td> <td></td> </tr> </table>				Name of Instructor & Department		1.	Dr Prashant Kumar, Assistant Professor, Computer Science and Engineering	2.	Dr OP Verma, Assistant Professor, Instrumentation and Control Engineering.	3.	
Name of Instructor & Department													
1.	Dr Prashant Kumar, Assistant Professor, Computer Science and Engineering												
2.	Dr OP Verma, Assistant Professor, Instrumentation and Control Engineering.												
3.													
10	Existing courses with overlapping content along with %age overlapping <i>(Offered by the same or other departments)</i>	6% overlap with CS-605 Advanced Machine Learning, MTech Data Science and Engineering, Computer Science and Engineering 4% overlap with CS-546 Deep Learning MTech in Information Security, Computer Science and Engineering											
11	Other departments to whom the course may be relevant (if any):												

The above proposal for a new course is found to be acceptable by (DUGC/ DPGC) in its meeting held on _____. The committee recommends this course proposal for consideration of SUGC / SPGC.

**Signature of the Chairman,
DUGC/ DPGC**

Proposal for New Academic Course

Name of the Centre: Centre for Artificial Intelligence

1	Title of the Course	Concurrent Programming Laboratory											
2	Credit Structure	<i>Lectures (L)</i>	<i>Tutorials (T)</i>	<i>Practicals (P)</i>	<i>Total Credits (C)</i>								
		0	0	3	2								
3	Academic Programme for which course is applicable	☐ Undergraduate ☒ Postgraduate ☐ Ph.D.											
4	Type of Course	☐ Theory ☐ Seminar ☒ Lab											
5	Course Tag	☒ Core ☐ Department Elective ☐ General Elective ☐ Minor											
6	Justification for Introduction of course	The <i>Concurrent Programming Laboratory</i> provides students with practical skills to design and implement parallel and concurrent systems, essential for high-performance computing and modern multi-core architectures. Through hands-on experience with threads, synchronization, and shared memory, students learn to optimize resource usage and improve execution efficiency. This lab reinforces theoretical concepts and prepares students for real-world applications in areas such as systems programming, AI, and cloud computing, where concurrency is fundamental.											
7	Contents of the course & Tentative Lecture Plan	Attached within Main Curriculum											
8	Texts and References (Minimum 5 – Maximum 8) (Complete name of Author/ Title/ Edition/ Publisher/ Volume, Web references/ e-references, year of publication, etc.)	1. Sanders J, Kandrot E. <i>CUDA by Example</i>. <u>Pearson Education</u>, 2010. 2. Kurk DB, Hwu WW. <i>Programming Massively Parallel Processors, A Hands-on Approach</i>. <u>Morgan Kaufmann</u>, 2010 3. <u>Shane C</u>. <i>CUDA Programming</i>. <u>Elsevier Science</u>. 2012 4. NVIDIA CUDA Toolkit Documentation 5. CUDA samples and SDKs from NVIDIA											
9	Names of Instructors (Require the names of at least two permanent faculty members of NITJ for core course and one permanent faculty member of NITJ for elective course.)	<table border="1" style="width: 100%; border-collapse: collapse;"> <tr> <th colspan="2" style="text-align: left;">Name of Instructor & Department</th> </tr> <tr> <td style="width: 5%; text-align: center;">1.</td> <td>Dr Prashant Kumar, Assistant Professor, Computer Science and Engineering</td> </tr> <tr> <td style="text-align: center;">2.</td> <td>Dr OP Verma, Assistant Professor, Instrumentation and Control Engineering.</td> </tr> <tr> <td style="text-align: center;">3.</td> <td></td> </tr> </table>				Name of Instructor & Department		1.	Dr Prashant Kumar, Assistant Professor, Computer Science and Engineering	2.	Dr OP Verma, Assistant Professor, Instrumentation and Control Engineering.	3.	
Name of Instructor & Department													
1.	Dr Prashant Kumar, Assistant Professor, Computer Science and Engineering												
2.	Dr OP Verma, Assistant Professor, Instrumentation and Control Engineering.												
3.													
10	Existing courses with overlapping content along with %age overlapping (Offered by the same or other departments)	No Overlap.											
11	Other departments to whom the course may be relevant (if any):												

The above proposal for a new course is found to be acceptable by (DUGC/ DPGC) in its meeting held on _____. The committee recommends this course proposal for consideration of SUGC / SPGC.

**Signature of the Chairman,
DUGC/ DPGC**

Proposal for New Academic Course

Name of the Centre: Centre for Artificial Intelligence

1	Title of the Course	Cloud Infrastructure for AI											
2	Credit Structure	<i>Lectures (L)</i>	<i>Tutorials (T)</i>	<i>Practicals (P)</i>	<i>Total Credits (C)</i>								
		3	0	0	3								
3	Academic Programme for which course is applicable	☐ Undergraduate ☒ Postgraduate ☐ Ph.D.											
4	Type of Course	☒ Theory ☐ Seminar ☐ Lab											
5	Course Tag	☐ Core ☒ Department Elective ☐ General Elective ☐ Minor											
6	Justification for Introduction of course	This course provides a foundational understanding of the intertwined fields of Data Science and Artificial Intelligence. It explores how data-driven insights are crucial for building intelligent systems and how AI techniques enable sophisticated analysis and decision-making from vast datasets. Students will learn the end-to-end data science pipeline, from data acquisition and preprocessing to exploratory analysis, model development using machine learning algorithms, and the deployment and interpretation of AI solutions.											
7	Contents of the course & Tentative Lecture Plan	Attached within Main Curriculum											
8	Texts and References (Minimum 5 – Maximum 8) <i>(Complete name of Author/ Title/ Edition/ Publisher/ Volume, Web references/ e-references, year of publication, etc.)</i>	1. Kleppmann M. Designing Data-Intensive Applications. O'Reilly Media, 1st Edition, 2017. 2. Karamitsos I, Graf J. Architecting Data and Machine Learning Platforms: A Guide to Unlocking the Full Potential of Your Data. O'Reilly Media, 1st Edition, 2022. 3. Negnevitsky M. Artificial Intelligence: A Guide to Intelligent Systems. Addison-Wesley, 3rd Edition, 2011. 4. Liebowitz J. Data Analytics and AI. Routledge, 1st Edition, 2020. 5. Crickard P. Data Engineering with Python: Work with Massive Datasets, Manage Data Pipelines, and Build AI Applications. Packt Publishing, 1st Edition, 2021.											
9	Names of Instructors <i>(Require the names of at least two permanent faculty members of NITJ for core course and one permanent faculty member of NITJ for elective course.</i>	<table border="1" style="width: 100%; border-collapse: collapse;"> <tr> <th colspan="2" style="text-align: left;">Name of Instructor & Department</th> </tr> <tr> <td style="width: 5%; text-align: center;">1.</td> <td>Dr Prashant Kumar, Assistant Professor, Computer science and Engineering.</td> </tr> <tr> <td style="text-align: center;">2.</td> <td>Dr Urvashi, Assistant Professor, Computer science and Engineering.</td> </tr> <tr> <td style="text-align: center;">3.</td> <td>Dr Nisha Chaurasiya, Assistant Professor, Information and Technology</td> </tr> </table>				Name of Instructor & Department		1.	Dr Prashant Kumar, Assistant Professor, Computer science and Engineering.	2.	Dr Urvashi, Assistant Professor, Computer science and Engineering.	3.	Dr Nisha Chaurasiya, Assistant Professor, Information and Technology
Name of Instructor & Department													
1.	Dr Prashant Kumar, Assistant Professor, Computer science and Engineering.												
2.	Dr Urvashi, Assistant Professor, Computer science and Engineering.												
3.	Dr Nisha Chaurasiya, Assistant Professor, Information and Technology												
10	Existing courses with overlapping content along with %age overlapping <i>(Offered by the same or other departments)</i>	25% Overlap CS-532 Cloud Computing & Communication MTech Computer Science and Engineering											
11	Other departments to whom the course may be relevant (if any):												

The above proposal for a new course is found to be acceptable by (DUGC/ DPGC) in its meeting held on _____. The committee recommends this course proposal for consideration of SUGC / SPGC.

**Signature of the Chairman,
DUGC/ DPGC**

Proposal for New Academic Course

Name of the Centre: Centre for Artificial Intelligence

1	Title of the Course	Cloud-Native AI Development											
2	Credit Structure	<i>Lectures (L)</i>	<i>Tutorials (T)</i>	<i>Practicals (P)</i>	<i>Total Credits (C)</i>								
		3	0	0	3								
3	Academic Programme for which course is applicable	☐ Undergraduate ☒ Postgraduate ☐ Ph.D.											
4	Type of Course	☒ Theory ☐ Seminar ☐ Lab											
5	Course Tag	☐ Core ☒ Department Elective ☐ General Elective ☐ Minor											
6	Justification for Introduction of course	This course explores the development of Artificial Intelligence applications using cloud-native paradigms. It delves into building scalable and resilient AI solutions by leveraging serverless computing, container orchestration, microservices architectures, and event-driven programming on cloud platforms.											
7	Contents of the course & Tentative Lecture Plan	Attached within Main Curriculum											
8	Texts and References (Minimum 5 – Maximum 8) (Complete name of Author/ Title/ Edition/ Publisher/ Volume, Web references/ e-references, year of publication, etc.)	1. Ajit Pratap Kundan (Author), Mainak Chakraborty. <i>Monitoring Cloud-Native Applications: Lead Agile Operations Confidently Using Open Source Software</i>. Academic Press, 2021. 2. David Bounds, Premkumar Rangarajan. <i>Cloud Native AI and Machine Learning on AWS</i>, bpb, 2023. 3. Adrian Gonzalez Sanchez. <i>Azure OpenAI Service for Cloud Native Applications: Designing, Planning, and Implementing Generative AI Solutions</i>. Shroff/O'Reilly, 2024. 4. Avinash Pamisetty. <i>Cloud-Native Intelligence and Scalable AI for Supply Chain, Food Services, and Financial Infrastructure</i>, Aqua Publications, 2025.											
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11	Other departments to whom the course may be relevant (if any):												

The above proposal for a new course is found to be acceptable by (DUGC/ DPGC) in its meeting held on _____. The committee recommends this course proposal for consideration of

SUGC / SPGC.

**Signature of the Chairman,
DUGC/ DPGC**

Proposal for New Academic Course

Name of the Centre: Centre for Artificial Intelligence

1	Title of the Course	MLOps on Cloud Platforms											
2	Credit Structure	<i>Lectures (L)</i>	<i>Tutorials (T)</i>	<i>Practicals (P)</i>	<i>Total Credits (C)</i>								
		3	0	0	3								
3	Academic Programme for which course is applicable	☐ Undergraduate ☒ Postgraduate ☐ Ph.D.											
4	Type of Course	☒ Theory ☐ Seminar ☐ Lab											
5	Course Tag	☐ Core ☒ Department Elective ☐ General Elective ☐ Minor											
6	Justification for Introduction of course	This course focuses on the principles and practices of MLOps (Machine Learning Operations) within cloud environments. Students will learn how to build, deploy, and manage the entire machine learning lifecycle, from data preparation and model training to deployment, monitoring, and retraining, leveraging cloud-native tools and services.											
7	Contents of the course & Tentative Lecture Plan	Attached within Main Curriculum											
8	Texts and References (Minimum 5 – Maximum 8) <i>(Complete name of Author/ Title/ Edition/ Publisher/ Volume, Web references/ e-references, year of publication, etc.)</i>	1. Mark Treveil, Nicolas Omont, Clément Stenac. <i>Introducing MLOps: How to Scale Machine Learning in the Enterprise</i> . Shroff/O'Reilly, 2020. 2. Noah Gift, Alfredo Deza. <i>Practical MLOps: Operationalizing Machine Learning Models</i> , Shroff/O'Reilly, 2021. 3. Emmanuel Raj. <i>Engineering MLOps</i> . Packt Publishing, 2021. 4. Saurabh D. Dorle. <i>Ultimate MLOps for Machine Learning Models: Use Real Case Studies to Efficiently Build, Deploy, and Scale Machine Learning Pipelines with MLOps</i> . Kindle Edition, 2024. 5. Yaron Haviv, Noah Gift. <i>Implementing MLOps in the Enterprise: A Production-First Approach</i> . Shroff/O'Reilly, 2023											
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SUGC / SPGC.

**Signature of the Chairman,
DUGC/ DPGC**

Proposal for New Academic Course

Name of the Centre: Centre for Artificial Intelligence

1	Title of the Course	MLOps on Cloud Platforms											
2	Credit Structure	<i>Lectures (L)</i>	<i>Tutorials (T)</i>	<i>Practicals (P)</i>	<i>Total Credits (C)</i>								
		3	0	0	3								
3	Academic Programme for which course is applicable	☐ Undergraduate ☒ Postgraduate ☐ Ph.D.											
4	Type of Course	☒ Theory ☐ Seminar ☐ Lab											
5	Course Tag	☐ Core ☒ Department Elective ☐ General Elective ☐ Minor											
6	Justification for Introduction of course	This course focuses on the principles and practices of MLOps (Machine Learning Operations) within cloud environments. Students will learn how to build, deploy, and manage the entire machine learning lifecycle, from data preparation and model training to deployment, monitoring, and retraining, leveraging cloud-native tools and services.											
7	Contents of the course & Tentative Lecture Plan	Attached within Main Curriculum											
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SUGC / SPGC.

**Signature of the Chairman,
DUGC/ DPGC**

Proposal for New Academic Course

Name of the Centre: Centre for Artificial Intelligence

1	Title of the Course	Advanced Data Management, Engineering, and Preprocessing for AI											
2	Credit Structure	<i>Lectures (L)</i>	<i>Tutorials (T)</i>	<i>Practicals (P)</i>	<i>Total Credits (C)</i>								
		3	0	0	3								
3	Academic Programme for which course is applicable	☐ Undergraduate ☒ Postgraduate ☐ Ph.D.											
4	Type of Course	☒ Theory ☐ Seminar ☐ Lab											
5	Course Tag	☐ Core ☒ Department Elective ☐ General Elective ☐ Minor											
6	Justification for Introduction of course	This course delves into advanced concepts of data acquisition, management, and rigorous preprocessing techniques essential for building robust AI systems. Leveraging students' existing statistical knowledge, it focuses on complex data quality issues, sophisticated imputation methods, comprehensive feature engineering strategies, and the architectural implications of various data storage solutions. The emphasis is on preparing diverse data types for high-performance AI/ML model training.											
7	Contents of the course & Tentative Lecture Plan	Attached within Main Curriculum											
8	Texts and References (Minimum 5 – Maximum 8) (Complete name of Author/ Title/ Edition/ Publisher/ Volume, Web references/ e-references, year of publication, etc.)	1. Kleppmann M. <i>Designing Data-Intensive Applications</i>. O'Reilly Media, 1st Edition, 2017. 2. Karamitsos I, Graf J. <i>Architecting Data and Machine Learning Platforms: A Guide to Unlocking the Full Potential of Your Data</i>. O'Reilly Media, 1st Edition, 2022. 3. Negnevitsky M. <i>Artificial Intelligence: A Guide to Intelligent Systems</i>. Addison-Wesley, 3rd Edition, 2011. 4. Liebowitz J. <i>Data Analytics and AI</i>. Routledge, 1st Edition, 2020. 5. Crickard P. <i>Data Engineering with Python: Work with Massive Datasets, Manage Data Pipelines, and Build AI Applications</i>. Packt Publishing, 1st Edition, 2021											
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11	Other departments to whom the course may be relevant (if any):												

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The above proposal for a new course is found to be acceptable by (DUGC/ DPGC) in its meeting held on _____. The committee recommends this course proposal for consideration of SUGC / SPGC.

**Signature of the Chairman,
DUGC/ DPGC**

Proposal for New Academic Course

Name of the Centre: Centre for Artificial Intelligence

1	Title of the Course	Advanced Machine Learning Models for Data Analytics											
2	Credit Structure	<table border="1" style="width: 100%; border-collapse: collapse;"> <tr> <td style="width: 25%; text-align: center;"><i>Lectures (L)</i></td> <td style="width: 25%; text-align: center;"><i>Tutorials (T)</i></td> <td style="width: 25%; text-align: center;"><i>Practicals (P)</i></td> <td style="width: 25%; text-align: center;"><i>Total Credits (C)</i></td> </tr> <tr> <td style="text-align: center;">3</td> <td style="text-align: center;">0</td> <td style="text-align: center;">0</td> <td style="text-align: center;">3</td> </tr> </table>				<i>Lectures (L)</i>	<i>Tutorials (T)</i>	<i>Practicals (P)</i>	<i>Total Credits (C)</i>	3	0	0	3
<i>Lectures (L)</i>	<i>Tutorials (T)</i>	<i>Practicals (P)</i>	<i>Total Credits (C)</i>										
3	0	0	3										
3	Academic Programme for which course is applicable	☐ Undergraduate ☒ Postgraduate ☐ Ph.D.											
4	Type of Course	☒ Theory ☐ Seminar ☐ Lab											
5	Course Tag	☐ Core ☒ Department Elective ☐ General Elective ☐ Minor											
6	Justification for Introduction of course	Building on core AI/ML knowledge and advanced data preparation skills, this course dives into sophisticated machine learning models crucial for complex data analytics. It explores advanced ensemble techniques, specialized regression and classification models, advanced clustering methods, and deep learning foundations, emphasizing their theoretical underpinnings, algorithmic nuances, and application in extracting deeper insights and making robust predictions from diverse datasets.											
7	Contents of the course & Tentative Lecture Plan	Attached within Main Curriculum											
8	Texts and References (Minimum 5 – Maximum 8) <i>(Complete name of Author/ Title/ Edition/ Publisher/ Volume, Web references/ e-references, year of publication, etc.)</i>	1. Goodfellow I, Bengio Y, Courville A. <i>Deep Learning</i> . MIT Press, 1st Edition, 2016. 2. Provost F, Fawcett T. <i>Data Science for Business</i> . O'Reilly Media, 1st Edition, 2013. 3. Kleppmann M. <i>Designing Data-Intensive Applications</i> . O'Reilly Media, 1st Edition, 2017. 4. Negnevitsky M. <i>Artificial Intelligence: A Guide to Intelligent Systems</i> . Addison-Wesley, 3rd Edition, 2011. 5. Chollet F. <i>Deep Learning with Python</i> . Manning Publications, 2nd Edition, 2021.											
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**Signature of the Chairman,
DUGC/ DPGC**

Proposal for New Academic Course

Name of the Centre: Centre for Artificial Intelligence

1	Title of the Course	Advanced AI for Unstructured Data and Strategic Analytics											
2	Credit Structure	<i>Lectures (L)</i>	<i>Tutorials (T)</i>	<i>Practicals (P)</i>	<i>Total Credits (C)</i>								
		3	0	0	3								
3	Academic Programme for which course is applicable	☐ Undergraduate ☒ Postgraduate ☐ Ph.D.											
4	Type of Course	☒ Theory ☐ Seminar ☐ Lab											
5	Course Tag	☐ Core ☒ Department Elective ☐ General Elective ☐ Minor											
6	Justification for Introduction of course	This course delves into advanced AI techniques, specifically focusing on deep learning architectures for processing and deriving insights from unstructured data—Natural Language (text) and Vision (images/videos). Students will gain a deep conceptual understanding of modern NLP and Computer Vision models, exploring their application in strategic data analytics, and how these AI models contribute to sophisticated decision support and complex problem-solving.											
7	Contents of the course & Tentative Lecture Plan	Attached within Main Curriculum											
8	Texts and References (Minimum 5 – Maximum 8) (Complete name of Author/ Title/ Edition/ Publisher/ Volume, Web references/ e-references, year of publication, etc.)	1. Kleppmann M. <i>Designing Data-Intensive Applications</i>. O'Reilly Media, 1st Edition, 2017. 2. Karamitsos I, Graf J. <i>Architecting Data and Machine Learning Platforms: A Guide to Unlocking the Full Potential of Your Data</i>. O'Reilly Media, 1st Edition, 2022. 3. Negnevitsky M. <i>Artificial Intelligence: A Guide to Intelligent Systems</i>. Addison-Wesley, 3rd Edition, 2011. 4. Liebowitz J. <i>Data Analytics and AI</i>. Routledge, 1st Edition, 2020. 5. Crickard P. <i>Data Engineering with Python: Work with Massive Datasets, Manage Data Pipelines, and Build AI Applications</i>. Packt Publishing, 1st Edition, 2021											
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**Signature of the Chairman,
DUGC/ DPGC**

Proposal for New Academic Course

Name of the Centre: Centre for Artificial Intelligence

1	Title of the Course	Scalable AI Systems, MLOps, and Cloud Architectures											
2	Credit Structure	<i>Lectures (L)</i>	<i>Tutorials (T)</i>	<i>Practicals (P)</i>	<i>Total Credits (C)</i>								
		3	0	0	3								
3	Academic Programme for which course is applicable	☐ Undergraduate ☒ Postgraduate ☐ Ph.D.											
4	Type of Course	☒ Theory ☐ Seminar ☐ Lab											
5	Course Tag	☐ Core ☒ Department Elective ☐ General Elective ☐ Minor											
6	Justification for Introduction of course	This capstone course integrates all previously learned advanced AI/ML and data science concepts with the practical challenges of deploying and managing AI systems at scale. It covers the architectural design of big data platforms, the comprehensive principles of Machine Learning Operations (MLOps) for the entire AI lifecycle, and the strategic leveraging of cloud-based AI services. The focus is on building, deploying, and maintaining robust, scalable, and ethical AI-driven data products in real-world production environments.											
7	Contents of the course & Tentative Lecture Plan	Attached within Main Curriculum											
8	Texts and References (Minimum 5 – Maximum 8) <i>(Complete name of Author/ Title/ Edition/ Publisher/ Volume, Web references/ e-references, year of publication, etc.)</i>	1. McMahon A. <i>Machine Learning Engineering with Python: Manage the Production Life Cycle of Machine Learning Models Using MLOps with Practical Examples</i>. Packt Publishing, 1st Edition, 2021. 2. Singh P, Karuparti A. <i>Generative AI for Cloud Solutions: Architect Modern AI LLMs in Secure, Scalable, and Ethical Cloud Environments</i>. Packt Publishing, 1st Edition, 2023. 3. Osipov C. <i>MLOps Engineering at Scale: Deploying PyTorch Models on AWS</i>. Manning Publications, 1st Edition, 2020. 4. Walsh B. <i>Productionizing AI: How to Deliver AI B2B Solutions with Cloud and Python</i>. Apress, 1st Edition, 2020. 5. Lakshmanan V. <i>Data Science on the Google Cloud Platform: Implementing End-to-End Real-Time Data Pipelines: From Ingest to Machine Learning</i>. O'Reilly Media, 1st Edition, 2022.											
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11	Other departments to whom the course may be relevant (if any):	
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**Signature of the Chairman,
DUGC/ DPGC**

Proposal for New Academic Course

Name of the Centre: Centre for Artificial Intelligence

1	Title of the Course	Generative AI Fundamentals											
2	Credit Structure	<i>Lectures (L)</i>	<i>Tutorials (T)</i>	<i>Practicals (P)</i>	<i>Total Credits (C)</i>								
		3	0	0	3								
3	Academic Programme for which course is applicable	☐ Undergraduate ☒ Postgraduate ☐ Ph.D.											
4	Type of Course	☒ Theory ☐ Seminar ☐ Lab											
5	Course Tag	☐ Core ☒ Department Elective ☐ General Elective ☐ Minor											
6	Justification for Introduction of course	The <i>Generative AI Fundamentals</i> course introduces the core concepts and techniques behind AI systems that can create content—text, images, audio, and more. Covering models like GANs, VAEs, and transformers, it lays the groundwork for understanding how machines generate realistic and meaningful outputs. As generative AI is rapidly transforming industries such as design, media, healthcare, and education, this elective prepares students to explore innovative applications and build intelligent, creative systems with real-world impact.											
7	Contents of the course & Tentative Lecture Plan	Attached within Main Curriculum											
8	Texts and References (Minimum 5 – Maximum 8) <i>(Complete name of Author/ Title/ Edition/ Publisher/ Volume, Web references/ e-references, year of publication, etc.)</i>	1. Foster D. <i>Generative Deep Learning</i>. O'Reilly Media 2008 2. <u>Babcock</u> J, <u>Bali</u> R. <i>Generative AI with Python and TensorFlow</i>. <u>Packt Publishing</u>, 2021 3. Goodfellow I, Bengio Y, Courville A. <i>Deep Learning</i>. MIT Press. 2008 4. Bishop C. <i>Pattern Recognition and Machine Learning</i>. Springer New York. 2011 5. Burkov A. <i>The Hundred-Page Machine Learning Book</i>. O'Reilly Media. 2006 6. Tunstall L, Werra LV, Wolf T. <i>Natural Language Processing with Transformers</i>. O'Reilly Media. 2009											
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11	Other departments to whom the course may be relevant (if any):												

The above proposal for a new course is found to be acceptable by (DUGC/ DPGC) in its meeting held on _____. The committee recommends this course proposal for consideration of SUGC / SPGC.

**Signature of the Chairman,
DUGC/ DPGC**

Proposal for New Academic Course

Name of the Centre: Centre for Artificial Intelligence

1	Title of the Course	Large Language Models											
2	Credit Structure	<i>Lectures (L)</i>	<i>Tutorials (T)</i>	<i>Practicals (P)</i>	<i>Total Credits (C)</i>								
		3	0	0	3								
3	Academic Programme for which course is applicable	☐ Undergraduate ☒ Postgraduate ☐ Ph.D.											
4	Type of Course	☒ Theory ☐ Seminar ☐ Lab											
5	Course Tag	☐ Core ☒ Department Elective ☐ General Elective ☐ Minor											
6	Justification for Introduction of course	Large Language Models (LLMs) like GPT and BERT are at the core of modern AI applications, enabling tasks such as text generation, summarization, translation, and chatbots. This course offers foundational knowledge of transformer architectures, fine-tuning, and practical usage of LLMs, along with ethical considerations like bias and safety. With growing demand across industries, the course prepares students to understand, apply, and responsibly deploy LLMs in real-world AI systems.											
7	Contents of the course & Tentative Lecture Plan	Attached within Main Curriculum											
8	Texts and References (Minimum 5 – Maximum 8) (Complete name of Author/ Title/ Edition/ Publisher/ Volume, Web references/ e-references, year of publication, etc.)	1. Magnus Ekman. <i>Learning Deep Learning: Theory and Practice of Neural Networks, Computer Vision, Natural Language Processing, and Transformers using TensorFlow</i>, Addison-Wesley 2021. 2. Anirudh Koul, Siddha Ganju. <i>Practical Generative AI: Tools and Applications Using ChatGPT and OpenAI APIs</i>, O'Reilly Media, 2023 3. Rajadhyax D. <i>Decoding GPT : An Intuitive Understanding of Large Language Models Generative AI Machine Learning and Neural Networks</i>. StoryMirror Infotech Pvt. Ltd 2024. 4. OpenAI Cookbook and Documentation (https://platform.openai.com/docs) 5. Google Research Papers on PaLM, BERT, and T5 6. Hugging Face Transformers Library Documentation											
9	Names of Instructors (Require the names of at least two permanent faculty members of NITJ for core course and one permanent faculty member of NITJ for elective course.	<table border="1" style="width: 100%; border-collapse: collapse;"> <tr> <td colspan="2" style="text-align: center;">Name of Instructor & Department</td> </tr> <tr> <td style="width: 5%; text-align: center;">1.</td><td>Dr Arun Khosla, Professor, Electronics and Communication Engineering.</td> </tr> <tr> <td style="text-align: center;">2.</td><td>Dr Urvashi, Assistant Professor, Computer Science and Engineering</td> </tr> <tr> <td style="text-align: center;">3.</td><td>Dr Prashant Kumar, Assistant Professor, Computer Science and Engineering</td> </tr> </table>				Name of Instructor & Department		1.	Dr Arun Khosla, Professor, Electronics and Communication Engineering.	2.	Dr Urvashi, Assistant Professor, Computer Science and Engineering	3.	Dr Prashant Kumar, Assistant Professor, Computer Science and Engineering
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1.	Dr Arun Khosla, Professor, Electronics and Communication Engineering.												
2.	Dr Urvashi, Assistant Professor, Computer Science and Engineering												
3.	Dr Prashant Kumar, Assistant Professor, Computer Science and Engineering												
10	Existing courses with overlapping content along with %age overlapping (Offered by the same or other departments)	5% overlap with CS-518 Natural Language Processing and Information Retrieval. 10% overlap with CS-632 Natural Language Processing Both offered by Computer Science and Engineering.											
11	Other departments to whom the course may be relevant (if any):												

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The above proposal for a new course is found to be acceptable by (DUGC/ DPGC) in its meeting held on _____. The committee recommends this course proposal for consideration of SUGC / SPGC.

**Signature of the Chairman,
DUGC/ DPGC**

Proposal for New Academic Course

Name of the Centre: Centre for Artificial Intelligence

1	Title of the Course	Applied Generative AI and Prompt Engineering											
2	Credit Structure	<table border="1" style="width: 100%; text-align: center;"> <tr> <th><i>Lectures (L)</i></th><th><i>Tutorials (T)</i></th><th><i>Practicals (P)</i></th><th><i>Total Credits (C)</i></th></tr> <tr> <td>3</td><td>0</td><td>0</td><td>3</td></tr> </table>				<i>Lectures (L)</i>	<i>Tutorials (T)</i>	<i>Practicals (P)</i>	<i>Total Credits (C)</i>	3	0	0	3
<i>Lectures (L)</i>	<i>Tutorials (T)</i>	<i>Practicals (P)</i>	<i>Total Credits (C)</i>										
3	0	0	3										
3	Academic Programme for which course is applicable	☐ Undergraduate ☒ Postgraduate ☐ Ph.D.											
4	Type of Course	☒ Theory ☐ Seminar ☐ Lab											
5	Course Tag	☐ Core ☒ Department Elective ☐ General Elective ☐ Minor											
6	Justification for Introduction of course	Generative AI is revolutionizing content creation, code generation, design, and decision support across industries. Tools like ChatGPT, DALL·E, Codex, and Stable Diffusion demonstrate the transformative potential of large-scale foundation models. However, leveraging these models effectively requires a deep understanding of prompt engineering, fine-tuning, and responsible usage. This course equips students with practical skills to harness generative AI models for real-world applications. It covers text, image, and code generation, alongside techniques like zero-shot, few-shot prompting, fine-tuning, and model evaluation. Ethical considerations, safety, and misuse mitigation are also addressed. Given the growing industry adoption of generative AI tools, this course prepares students for modern AI roles by combining hands-on experience with a strategic understanding of model behavior and responsible deployment.											
7	Contents of the course & Tentative Lecture Plan	Attached within Main Curriculum											
8	Texts and References (Minimum 5 – Maximum 8) <i>(Complete name of Author/ Title/ Edition/ Publisher/ Volume, Web references/ e-references, year of publication, etc.)</i>	1. Sansevier O, Cuenca P, Passos A, Whitaker J. <i>Hands-On Generative AI with Transformers and Diffusion Models</i>. O'Reilly Media, 2012 2. Kharwal A. <i>From ML Algorithms to GenAI & LLMs: Master ML Algorithms and Generative AI & LLMs with Python from scratch</i>. Bluerose Publishers Pvt. Ltd. 2024. 3. Jha SK. <i>GenAI Foundations: LLMS From Fundamental to Future Frontiers</i>. Notion Press 2025. 4. Sabesan K, Sivagamisundari, Dutta N. <i>Generative AI for Everyone: Deep learning, NLP, and LLMs for creative and practical applications</i>. Bpb Publications 2025. 5. Ozdimer S. <i>Quick Start Guide to Large Language Models</i>. Addison-Wesley Professional 2024. 6. Pai S. <i>Designing Large Language Model Applications: A Holistic Approach to LLMs</i>. O'Reilly Media 2025.											

9	Names of Instructors <i>(Require the names of at least two permanent faculty members of NITJ for core course and one permanent faculty member of NITJ for elective course.</i>	<table><tr><th colspan="2">Name of Instructor & Department</th></tr><tr><td>1.</td><td>Dr Arun Khosla, Professor, Electronics and Communication Engineering.</td></tr><tr><td>2.</td><td>Dr OP Verma, Assistant Professor</td></tr><tr><td>3.</td><td>Dr Prashant Kumar, Assistant Professor, Computer Science and Engineering</td></tr></table>	Name of Instructor & Department		1.	Dr Arun Khosla, Professor, Electronics and Communication Engineering.	2.	Dr OP Verma, Assistant Professor	3.	Dr Prashant Kumar, Assistant Professor, Computer Science and Engineering
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2.	Dr OP Verma, Assistant Professor									
3.	Dr Prashant Kumar, Assistant Professor, Computer Science and Engineering									
10	Existing courses with overlapping content along with %age overlapping <i>(Offered by the same or other departments)</i>	No Overlap.								
11	Other departments to whom the course may be relevant (if any):									

The above proposal for a new course is found to be acceptable by (DUGC/ DPGC) in its meeting held on _____. The committee recommends this course proposal for consideration of SUGC / SPGC.

Signature of the Chairman,
DUGC/ DPGC

Proposal for New Academic Course

Name of the Centre: Centre for Artificial Intelligence

1	Title of the Course	Agentic AI											
2	Credit Structure	<i>Lectures (L)</i>	<i>Tutorials (T)</i>	<i>Practicals (P)</i>	<i>Total Credits (C)</i>								
		3	0	0	3								
3	Academic Programme for which course is applicable	☐ Undergraduate ☒ Postgraduate ☐ Ph.D.											
4	Type of Course	☒ Theory ☐ Seminar ☐ Lab											
5	Course Tag	☐ Core ☒ Department Elective ☐ General Elective ☐ Minor											
6	Justification for Introduction of course	Agentic AI focuses on autonomous, goal-driven systems capable of reasoning, planning, and taking actions in dynamic environments. From intelligent virtual assistants to autonomous robotics and multi-agent coordination, these systems represent the next evolution of AI beyond passive prediction. This course equips students with foundational concepts in agent design, decision-making, reinforcement learning, and multi-agent interactions. With increasing adoption in fields like robotics, gaming, finance, and logistics, this elective prepares students to build intelligent agents capable of adaptive and autonomous behavior in real-world settings.											
7	Contents of the course & Tentative Lecture Plan	Attached within Main Curriculum											
8	Texts and References (Minimum 5 – Maximum 8) (Complete name of Author/ Title/ Edition/ Publisher/ Volume, Web references/ e-references, year of publication, etc.)	1. David Foster. <i>Generative Deep Learning: Teaching Machines to Paint, Write, Compose, and Play</i>, O'Reilly Media 2023 2. Magnus Ekman. <i>Learning Deep Learning: Theory and Practice of Neural Networks, Computer Vision, Natural Language Processing, and Transformers using TensorFlow</i>, Addison-Wesley 2021). 3. Josh Kalin. <i>Generative Adversarial Networks Cookbook: Over 100 recipes to build generative models using Python, TensorFlow, and Keras</i>. Packt Publishing. 2021. 4. Anirudh Koul, Siddha Ganju. <i>Practical Generative AI: Tools and Applications Using ChatGPT and OpenAI APIs</i>, O'Reilly Media, 2023											
9	Names of Instructors (Require the names of at least two permanent faculty members of NITJ for core course and one permanent faculty member of NITJ for elective course.	<table border="1" style="width: 100%; border-collapse: collapse;"> <tr> <td colspan="2" style="text-align: center;">Name of Instructor & Department</td> </tr> <tr> <td style="width: 5%; text-align: center;">1.</td> <td>Dr Arun Khosla, Professor, Electronics and Communication Engineering.</td> </tr> <tr> <td style="text-align: center;">2.</td> <td>Dr Urvashi, Assistant Professor, Computer Science and Engineering</td> </tr> <tr> <td style="text-align: center;">3.</td> <td>Dr Prashant Kumar, Assistant Professor, Computer Science and Engineering</td> </tr> </table>				Name of Instructor & Department		1.	Dr Arun Khosla, Professor, Electronics and Communication Engineering.	2.	Dr Urvashi, Assistant Professor, Computer Science and Engineering	3.	Dr Prashant Kumar, Assistant Professor, Computer Science and Engineering
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10	Existing courses with overlapping content along with %age overlapping (Offered by the same or other departments)	No Overlap											
11	Other departments to whom the												

	course may be relevant (if any):	
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The above proposal for a new course is found to be acceptable by (DUGC/ DPGC) in its meeting held on _____. The committee recommends this course proposal for consideration of SUGC / SPGC.

**Signature of the Chairman,
DUGC/ DPGC**

Proposal for New Academic Course

Name of the Centre: Centre for Artificial Intelligence

1	Title of the Course	Fundamentals of Signal and Image Processing											
2	Credit Structure	<i>Lectures (L)</i>	<i>Tutorials (T)</i>	<i>Practicals (P)</i>	<i>Total Credits (C)</i>								
		3	0	0	3								
3	Academic Programme for which course is applicable	☐ Undergraduate ☒ Postgraduate ☐ Ph.D.											
4	Type of Course	☒ Theory ☐ Seminar ☐ Lab											
5	Course Tag	☐ Core ☒ Department Elective ☐ General Elective ☐ Minor											
6	Justification for Introduction of course	Signal and image processing form the foundation for a wide range of AI applications, including speech recognition, medical imaging, remote sensing, and computer vision. This course introduces core concepts such as filtering, Fourier transforms, convolution, and edge detection, enabling students to effectively analyze and preprocess real-world data. It provides essential theoretical and practical knowledge that supports advanced work in AI, making it a valuable elective for students aiming to specialize in vision, audio, or sensor-based intelligent systems.											
7	Contents of the course & Tentative Lecture Plan	Attached within Main Curriculum											
8	Texts and References (Minimum 5 – Maximum 8) <i>(Complete name of Author/ Title/ Edition/ Publisher/ Volume, Web references/ e-references, year of publication, etc.)</i>	1. <u>Oppenheim AV, Verghese GC. Signals, Systems and Inference. Pearson Education, 2018.</u> 2. <u>Proakis JG. Digital Signal Processing: Principles, Algorithms and Applications. Prentice Hall, 2007</u> 3. <u>Gonzalez RC, Woods RE. Digital Image Processing. Pearson. 2018</u> 4. <u>Downey AB. Think DSP: Digital Signal Processing in Python. O'Reilly Media, 2016.</u>											
9	Names of Instructors <i>(Require the names of at least two permanent faculty members of NITJ for core course and one permanent faculty member of NITJ for elective course.</i>	<table border="1" style="width: 100%; border-collapse: collapse;"> <tr> <th colspan="2" style="text-align: left;">Name of Instructor & Department</th> </tr> <tr> <td style="width: 5%; text-align: center;">1.</td> <td>Dr Arun Khosla, Professor, Electronics and Communication Engineering.</td> </tr> <tr> <td style="text-align: center;">2.</td> <td>Dr OP Verma, Assistant Professor, Instrumentation and Control Engineering</td> </tr> <tr> <td style="text-align: center;">3.</td> <td>Dr KC Sharma, Assistant Professor, Electrical Engineering</td> </tr> </table>				Name of Instructor & Department		1.	Dr Arun Khosla, Professor, Electronics and Communication Engineering.	2.	Dr OP Verma, Assistant Professor, Instrumentation and Control Engineering	3.	Dr KC Sharma, Assistant Professor, Electrical Engineering
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1.	Dr Arun Khosla, Professor, Electronics and Communication Engineering.												
2.	Dr OP Verma, Assistant Professor, Instrumentation and Control Engineering												
3.	Dr KC Sharma, Assistant Professor, Electrical Engineering												
10	Existing courses with overlapping content along with %age overlapping <i>(Offered by the same or other departments)</i>	20% overlap with EC 519 Advanced Signal and Image Processing in MTech ECE, Electronics and communication engineering.											
11	Other departments to whom the course may be relevant (if any):												

The above proposal for a new course is found to be acceptable by (DUGC/ DPGC) in its meeting held on _____. The committee recommends this course proposal for consideration of SUGC / SPGC.

**Signature of the Chairman,
DUGC/ DPGC**

Proposal for New Academic Course

Name of the Centre: Centre for Artificial Intelligence

1	Title of the Course	Deep Learning for Signal and Image Processing											
2	Credit Structure	<i>Lectures (L)</i>	<i>Tutorials (T)</i>	<i>Practicals (P)</i>	<i>Total Credits (C)</i>								
		3	0	0	3								
3	Academic Programme for which course is applicable	☐ Undergraduate ☒ Postgraduate ☐ Ph.D.											
4	Type of Course	☒ Theory ☐ Seminar ☐ Lab											
5	Course Tag	☐ Core ☒ Department Elective ☐ General Elective ☐ Minor											
6	Justification for Introduction of course	Deep learning has significantly advanced the capabilities of signal and image processing, enabling breakthroughs in object recognition, speech processing, biomedical imaging, and more. This course focuses on applying CNNs, RNNs, and transformer-based models to complex visual and temporal data. It equips students with hands-on experience in building deep learning pipelines for classification, segmentation, enhancement, and denoising tasks—bridging foundational processing techniques with modern AI solutions. This elective is ideal for students aiming to work in fields like computer vision, audio analytics, and medical diagnostics.											
7	Contents of the course & Tentative Lecture Plan	Attached within Main Curriculum											
8	Texts and References (Minimum 5 – Maximum 8) <i>(Complete name of Author/ Title/ Edition/ Publisher/ Volume, Web references/ e-references, year of publication, etc.)</i>	1. Elgendy M. <i>Deep Learning for Vision Systems</i>. <u>Manning</u>, 2020 2. Solem JE. <i>Programming Computer Vision with Python: Tools and Algorithms for Analyzing Images</i>. <u>O'Reilly Media</u>, 2012 3. Camastra F, Vinciarelli A. <i>Machine Learning for Audio, Image and Video Analysis</i>, Springer Nature. 2015. 4. Ma Y, Qian G. <i>Intelligent Video Surveillance: Systems and Technology</i>, CRC Press, 2009. 5. Nilsson F, Axis C. <i>Intelligent Network Video: Understanding Modern Video Surveillance Systems</i>, CRC Press. 2017.											
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10	Existing courses with overlapping content along with %age overlapping <i>(Offered by the same or other departments)</i>	No Overlap											
11	Other departments to whom the course may be relevant (if any):												

The above proposal for a new course is found to be acceptable by (DUGC/ DPGC) in its meeting held on _____. The committee recommends this course proposal for consideration of SUGC / SPGC.

**Signature of the Chairman,
DUGC/ DPGC**

Proposal for New Academic Course

Name of the Centre: Centre for Artificial Intelligence

1	Title of the Course	Machine Learning for Signal and Image Processing											
2	Credit Structure	<i>Lectures (L)</i>	<i>Tutorials (T)</i>	<i>Practicals (P)</i>	<i>Total Credits (C)</i>								
		3	0	0	3								
3	Academic Programme for which course is applicable	☐ Undergraduate ☒ Postgraduate ☐ Ph.D.											
4	Type of Course	☒ Theory ☐ Seminar ☐ Lab											
5	Course Tag	☐ Core ☒ Department Elective ☐ General Elective ☐ Minor											
6	Justification for Introduction of course	Modern AI applications in healthcare, surveillance, multimedia, and remote sensing increasingly rely on machine learning to analyze complex signals and images. This course bridges foundational signal/image processing with ML techniques, covering classification, segmentation, denoising, and feature extraction using supervised and unsupervised learning. It equips students with practical skills to build intelligent systems that can learn from visual and sensory data, making it a crucial elective for those pursuing careers in computer vision, biomedical AI, or embedded intelligent systems..											
7	Contents of the course & Tentative Lecture Plan	Attached within Main Curriculum											
8	Texts and References (Minimum 5 – Maximum 8) <i>(Complete name of Author/ Title/ Edition/ Publisher/ Volume, Web references/ e-references, year of publication, etc.)</i>	1. <u>Brownlee J.</u> <i>Introduction to Time Series Forecasting with Python.</i> Jason Brownlee. 2017. 2. <u>Murohy KP.</u> <i>Machine Learning: A Probabilistic Perspective.</i> MIT Press, 2012 3. <u>Rangayyan RM.</u> <i>Biomedical Signal Analysis.</i> Wiley. 2015 4. <u>Downey AB.</u> <i>Think DSP: Digital Signal Processing in Python.</i> O'Reilly Media, 2016. 5. Scikit-learn and OpenCV documentation											
9	Names of Instructors <i>(Require the names of at least two permanent faculty members of NITJ for core course and one permanent faculty member of NITJ for elective course.</i>	<table border="1" style="width: 100%; border-collapse: collapse;"> <tr> <th colspan="2" style="text-align: left;">Name of Instructor & Department</th> </tr> <tr> <td style="width: 5%; text-align: center;">1.</td> <td>Dr Arun Khosla, Professor, Electronics and Communication Engineering.</td> </tr> <tr> <td style="text-align: center;">2.</td> <td>Dr OP Verma, Assistant Professor, Instrumentation and Control Engineering</td> </tr> <tr> <td style="text-align: center;">3.</td> <td>Dr KC Sharma, Assistant Professor, Electrical Engineering</td> </tr> </table>				Name of Instructor & Department		1.	Dr Arun Khosla, Professor, Electronics and Communication Engineering.	2.	Dr OP Verma, Assistant Professor, Instrumentation and Control Engineering	3.	Dr KC Sharma, Assistant Professor, Electrical Engineering
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10	Existing courses with overlapping content along with %age overlapping <i>(Offered by the same or other departments)</i>	No Overlap											
11	Other departments to whom the course may be relevant (if any):												

The above proposal for a new course is found to be acceptable by (DUGC/ DPGC) in its meeting held on _____. The committee recommends this course proposal for consideration of SUGC / SPGC.

**Signature of the Chairman,
DUGC/ DPGC**

Proposal for New Academic Course

Name of the Centre: Centre for Artificial Intelligence

1	Title of the Course	Deployable AI applications in Signals and Image Processing											
2	Credit Structure	<i>Lectures (L)</i>	<i>Tutorials (T)</i>	<i>Practicals (P)</i>	<i>Total Credits (C)</i>								
		3	0	0	3								
3	Academic Programme for which course is applicable	☐ Undergraduate ☒ Postgraduate ☐ Ph.D.											
4	Type of Course	☒ Theory ☐ Seminar ☐ Lab											
5	Course Tag	☐ Core ☒ Department Elective ☐ General Elective ☐ Minor											
6	Justification for Introduction of course	AI techniques are increasingly being tailored for domain-specific challenges in time series (e.g., forecasting, anomaly detection) and image analysis (e.g., segmentation, object detection). This course focuses on applying and deploying advanced AI models in sectors like healthcare, finance, remote sensing, and industrial automation. Students will gain hands-on experience in real-world data scenarios, learn deployment practices using edge/cloud platforms, and understand the constraints of operational AI systems. The course bridges advanced analytics with practical deployment, preparing students for AI roles in high-impact domains.											
7	Contents of the course & Tentative Lecture Plan	Attached within Main Curriculum											
8	Texts and References (Minimum 5 – Maximum 8) <i>(Complete name of Author/ Title/ Edition/ Publisher/ Volume, Web references/ e-references, year of publication, etc.)</i>	1. <u>Brownlee J.</u> <i>Introduction to Time Series Forecasting with Python.</i> Jason Brownlee. 2017 2. <u>Monlar C.</u> <i>Interpretable Machine Learning: A Guide for Making Black Box Models Explainable.</i> Leanpub. 2020. 3. Shen D, Greenspan H, Zhou SK. <i>Deep Learning for Medical Image Analysis.</i> Elsevier Science, 2017 4. Louk A, Ganju S, Kasam M. <i>Practical Deep Learning for Cloud, Mobile, and Edge: Real-World AI & Computer-Vision Projects Using Python, Keras & TensorFlow.</i> O'Reilly Media. 2019.											
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10	Existing courses with overlapping content along with %age overlapping <i>(Offered by the same or other departments)</i>	No Overlap											
11	Other departments to whom the course may be relevant (if any):												

The above proposal for a new course is found to be acceptable by (DUGC/ DPGC) in its meeting held on _____. The committee recommends this course proposal for consideration of SUGC / SPGC.

**Signature of the Chairman,
DUGC/ DPGC**